COLUMBIA

#StatFunGen

# Gene mapping in complex trait functional genomics

## From single markers to systems biology

Gao Wang, PhD, Assistant Professor of Neurological Sciences

January 29, 2025

1. Overview of post-GWAS analysis
2. Statistical fine-mapping in GWAS and QTL studies
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4. Overview of molecular quantitative trait loci (QTL) studies
5. QTL-GWAS loci: multi-trait analysis and colocalization
6. QTL-GWAS genes: transcriptome-wide association studies
7. Exercise: multivariate fine-mapping and TWAS
8. Connections: fine-mapping, colocalization, TWAS and MR

# Overview of post-GWAS analysis

# Follow-up questions on GWAS

**Statistical**

- How many independent association signals exist?
- Which variants are likely causal vs. tagging?

**Biological**

- Which genes mediate these associations?
- Through which biological pathways?
- In which tissues/cell-types do these variants act?

**Translational**

- Can we identify promising drug targets?
- Can we predict disease risk or trait values?
- How can we prioritize follow-up experiments?

# Sources of association signals

Causal association — meaningful

- Tested genetic variations influence traits directly

Linkage disequilibrium (LD) — useful

- Tested genetic variations associated with other nearby variations that influence traits
- Meaningful or misleading, in different contexts

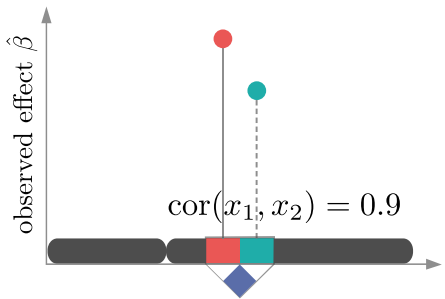
Population stratification — misleading

- Tested genetic variations is unrelated to traits, but is associated due to sampling differences
- eg, minor allele frequency, disease prevalence

# Impact of LD on GWAS analysis

**Oligogenic**: trait influenced by a few genetic variants

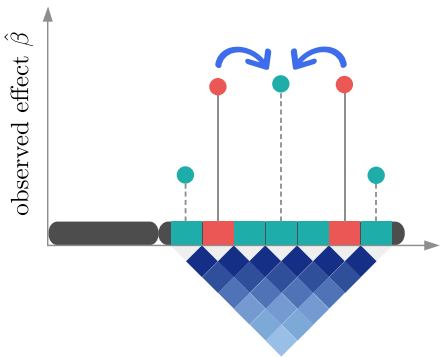
- Misleading: difficult to identify causal variants
- Useful: ‘tag SNPs’ in array based GWAS design



# Impact of LD on GWAS analysis

**Polygenic**: trait influenced by numerous genetic variants

- Misleading: spurious strong association due to LD ‘buddies’
- Useful: whole-genome prediction with sparse models



# Post-GWAS analysis toolkit

**Variant-level analysis**

- Fine-mapping: identify likely causal variants
- Colocalization: shared genetic effects
- Mendelian randomization: causal inference

**Aggregated analysis**

- LD score regression: confounding, heritability, functional enrichment and genetic correlation
- Gene-based tests and pathway enrichment
- Transcriptome-wide association (TWAS)
- Polygenic prediction

**Key challenge:** Connecting statistical associations to biological mechanisms

Kontou and Bagos *BioData Mining* (2024) 17:31
https://doi.org/10.1186/s13040-024-00385-x

BioData Mining

RESEARCH Open Access

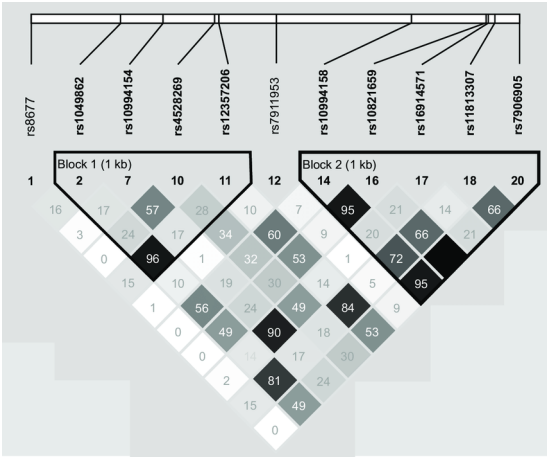
# The goldmine of GWAS summary statistics: a systematic review of methods and tools

Panagiota I. Kontou[1] and Pantelis G. Bagos[2*]

**Figure:** "Our review identified a total of 305 functioning software tools and databases dedicated to GWAS summary statistics, each with unique strengths and limitations."

# LD and LD score regression

# How many independent associations?

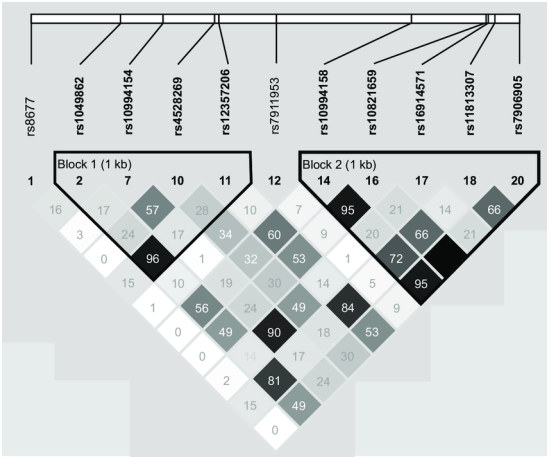



**LD clumping**

- Select lead SNP by p-value
- Remove SNPs in LD given fixed threshold (*e.g.* $r^2 > 0.1$)
- Within fixed window (*e.g.* 1Mb)
- Implementation: PLINK, bigsnpr

Note: LD pruning can be implemented as LD clumping on MAF.

# How many independent associations?

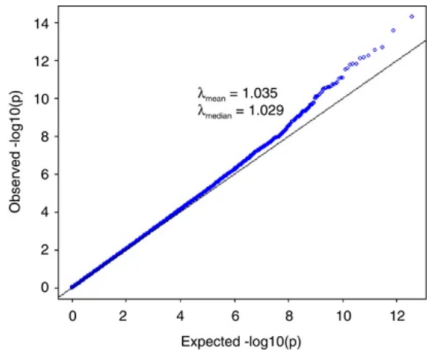



**Limitations**

- **Population-specific:** Window size and LD patterns vary by ancestry
- **Strong effects bias:**
  - Even very low LD SNPs near well-powered loci can show high significance
  - Example: APOE4 allele in Alzheimer's disease GWAS drives signals across chromosome 19

# A second thought on genomic inflation

Population stratification? Or, polygenic inheritance + LD?




Suggested reading: Yang et al (2011) EJHG

# LD score regression

**Initial motivation:** distinguish polygenicity from confounding

- Under pure genetic drift, LD is uncorrelated to magnitude of allele frequency differences between populations

# LD score regression

**Initial motivation:** distinguish polygenicity from confounding

- Under pure genetic drift, LD is uncorrelated to magnitude of allele frequency differences between populations

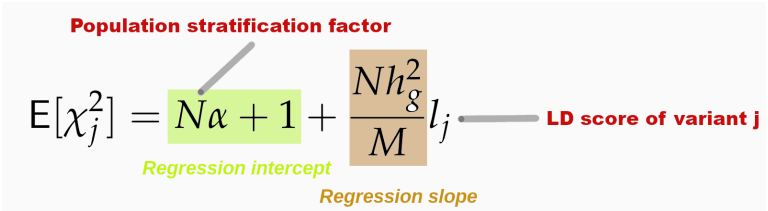
Assuming each SNP explains the same amount of trait variance,

$$E[\chi_j^2|\ell_j] = 1 + Na + \frac{Nh^2}{M}\ell_j$$

- **LD score of SNP j:** $\ell_j = \sum_{k=1}^{M} r_{jk}^2$
- $N$ is the GWAS sample size
- $\frac{h^2}{M}$ is the average heritability explained per SNP
- $a \neq 0$ indicates **confounding** (*e.g.* population stratification).

# LD score regression (LDSC)

Separating $h_g^2$ and population stratification




A more powerful and accurate correction factor for GWAS summary statistics compared to genomic control approach.

- Bulik-Sullivan et al (2015) Nature Genetics — the LDSC regression paper
- Zhu and Stephens (2017) AoAS — a neat, alternative LDSC regression model derivation in supplemental material

# LDSC application: heritability estimation

Narrow sense heritibility

- Proportion of phenotypic variation explained by additive genetic factors

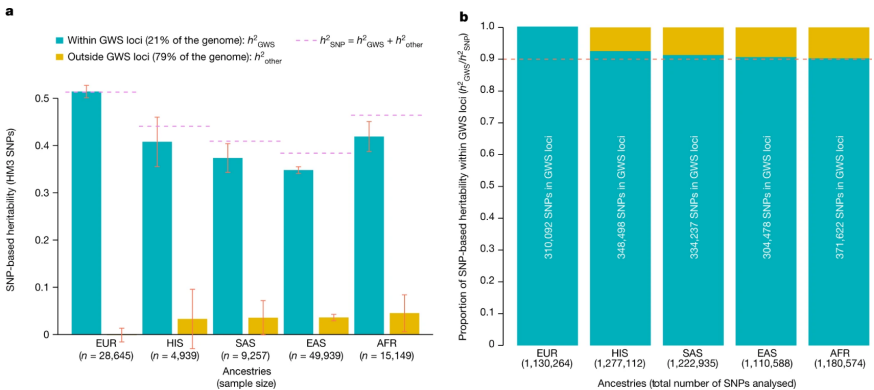
Estimation strategy

- Pedigree design: genetic covariance and IBD sharing
- Population design: linear mixed models

Population design, summary statistics

- LDSC to estimate SNP-based heritability
- Stratified LDSC (S-LDSC) to partition heritability by functional annotations

# Variance of height explained in GWAS




Yengo *et al*. (2022) Nature

# Functional annotation and enrichment analysis

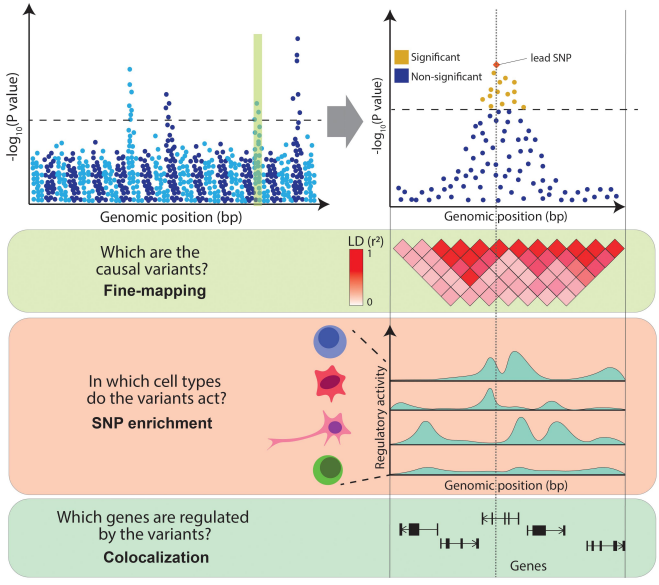
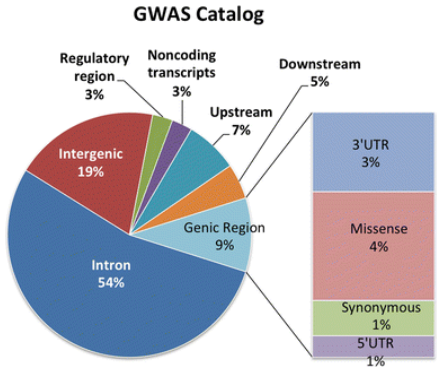



**Figure:** Cano-Gamez and Trynka (2020) Front. Genet.

# GWAS variants catelog by functional annotations

Most GWAS associations are non-coding




Lee *et al.* (2018) *Human Genetics*

# Enrichment of functional annotations in GWAS

**Question:** where in the genome is heritability concentrated?

- Integrate directly as range based binary annotations
  - Finucane et al (2015) Nature Genetics — Stratified LDSC paper
- Extension: variant specific continuous annotations
  - Gazal et al (2017) Nature Genetics
- Tissue specific variant level annotations independent of GWAS results
  - Deep Learning methods
  - Zhou et al (2015) Nature Genetics, Zhou et al (2018) Nature Genetics, Lai *et al*. (2022) PLoS Comp Bio
  - Avsec *et al*. (2021) Nature Methods

# A Polygenic Model: Partitioned Heritability

Stratified LD Score Regression assumes heritability varies linearly with functional annotations:

$$E[\chi_j^2 | \ell_{jA_1}, \ldots, \ell_{jA_K}] = 1 + Na + N \sum_{k=1}^{K} \tau_k \ell_{jA_k}$$

- $\tau_k = \frac{h_{gk}^2}{M_k}$ quantifies heritability per SNP within annotation $A_k$.
- $\ell_{jA_k} = \sum w_k r_{jk}^2$ is weighted LD score for SNP $j$ based on SNPs within annotation $A_k$.
- $w_k$ can be binary (e.g., 0/1) or continuous (e.g., functional relevance scores).
- **Enrichment:** relative contribution of heritability explained by annotation $A_k$, $\frac{\tau_k}{\frac{h_g^2}{M}}$.

# S-LDSC is a genome-wide approach

- A single locus may not have enough power to leverage annotation enrichment
- Genome-wide evaluation of thousands of annotations can increase power of fine-mapping
  - Lead to new loci to discover
- Functional enrichment can be done under the same framework as heritability estimation
  - Prioritize genomic features / tissues / cell-types
- **Enrichment coefficient may be transferrable cross population**
  - Weissbrod *et al*. (2022) Nat. Genet.

# Cell-type enrichment in GWAS traits via S-LDSC

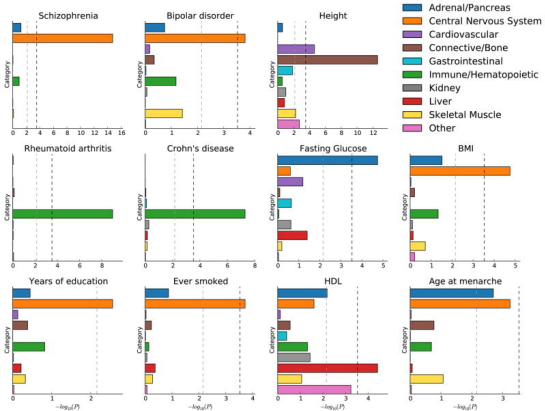



**Figure:** Finucane *et al*. (2015) Nature Genetics

# A sparse model (a somewhat oligogenic view)

Generalized linear model for SNP effects given $K$ annotations

$$\beta_j = (1 - \pi_j)\delta_0 + \pi_j g(\Theta)$$

$$\pi_j := \Pr(\gamma_j = 1 | \boldsymbol{\alpha}, \boldsymbol{d})$$

$$\log\left[\frac{\pi_j}{1 - \pi_j}\right] = \alpha_0 + \sum_{k=1}^{K} \alpha_k d_{kj}$$

$\alpha$ are **log fold enrichment** of functional genomic features

- Suggested reading: Wen (2016) AoAS

# Enrichment of DNase I in GTEx eQTLs

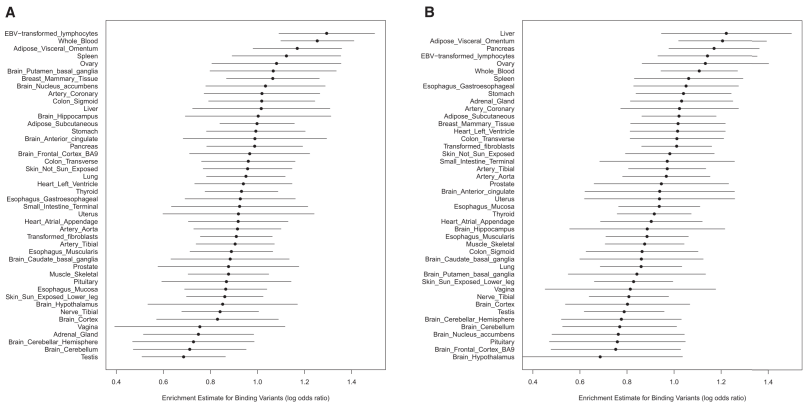



**Figure:** Wen *et al*. (2016) AJHG

# Gene Sets / Pathway Databases

**Gene Ontology (GO):**

- Three principal categories: Biological Processes, Cellular Components and Molecular Functions.
- GO terms have hierarchical structure, from broad to narrow (e.g., metabolism → GTP biosynthesis).
- Each GO term is linked to a curated list of associated genes.

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**KEGG:**

- Focuses on metabolic, disease, and trait-related pathways.
- Smaller and more curated compared to GO, emphasizing enzymatic and biological networks compared to disease and trait related.

# Over-Representation Analysis for Functional Enrichment

| | In Pathway | Not in Pathway |
|---|---|---|
| **Association Signal** | a | b |
| **No Association Signal** | c | d |

**Statistical tests:** Fisher's Exact Test, $\chi^2$-test, Binomial proportion test.

# Over-Representation Analysis for Functional Enrichment

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| **Association Signal** | a | b |
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**Statistical tests:** Fisher's Exact Test, $\chi^2$-test, Binomial proportion test.

**Jackknife standard error:** $\text{SE}_{\text{jack}} = \sqrt{\frac{N-1}{N}\sum_{i=1}^{N}(\hat{\theta}_{-i} - \bar{\theta}_{(\cdot)})^2}$

- $\hat{\theta}_{-i}$ is estimate with *i*th chromosome removed, $\bar{\theta}_{(\cdot)}$ is mean of leave-one-out estimates, *N* is number of chromosomes
- A convenient way to account for dependent genetic variables (LD) and annotations (*e.g.* overlapping gene-sets)

# Over-Representation Analysis for Functional Enrichment

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|---|---|---|
| **Association Signal** | a | b |
| **No Association Signal** | c | d |

**Statistical tests:** Fisher's Exact Test, $\chi^2$-test, Binomial proportion test.

**Remaining sources of bias:**

- LD proxy / buddies (e.g., captured by LD score).
- Gene density (longer genes capture more SNPs).
- Distance to transcription start site (TSS).
- Minor allele frequency (MAF).

**SNPs for ORA can be chosen to match on these factors to eliminate bias.**

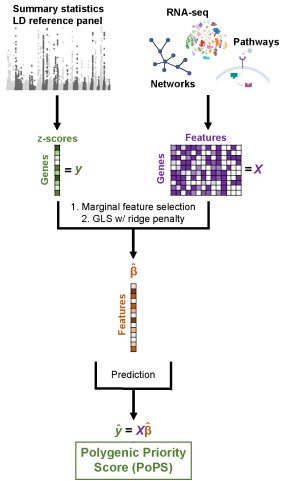
# Other Enrichment Analysis Methods

**Additional approaches:**

- Gene Set Enrichment Analysis (GSEA)
- Stratified Linkage Disequilibrium Score Regression (sLDSC)

**Polygenic Priority Score (PoPS):**

- An extension to sLDSC and GSEA.
- Integrates polygenic signals across gene sets.
- **Reference:** Weeks et al. (2023) *Nature Genetics*.



# Statistical fine-mapping in GWAS and QTL studies

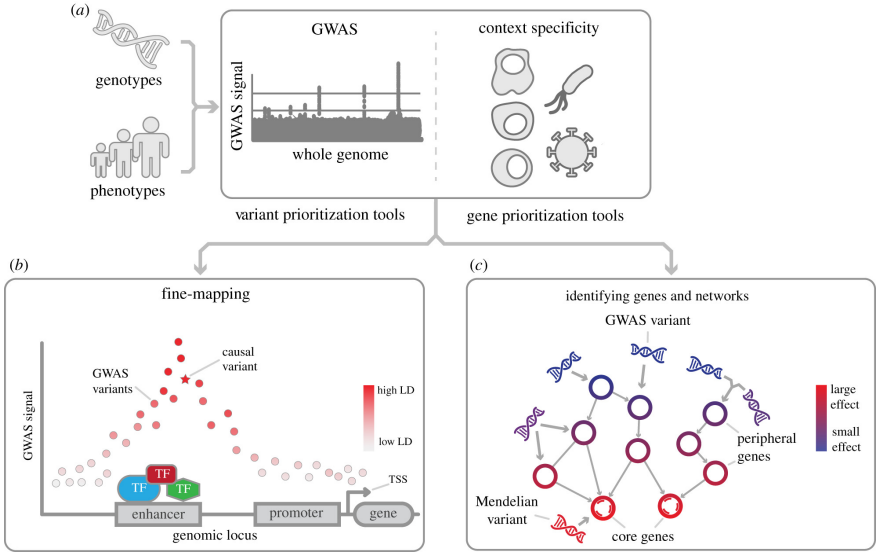
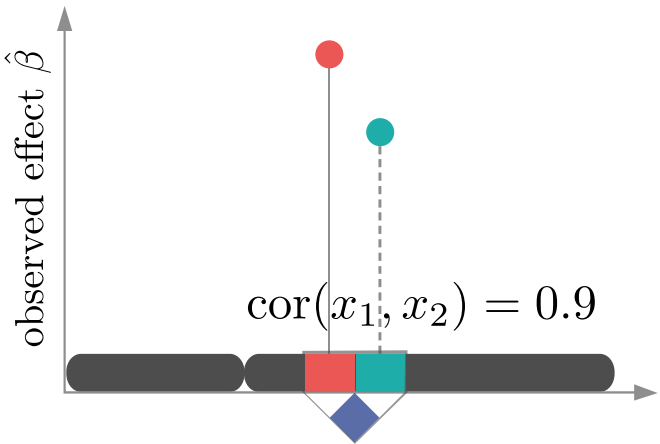



**Figure:** Broekema *et al*. (2020) Open Biol.

# Correlated variables in association studies

Due to a phenomenon called **linkage disequilibrium** (LD)



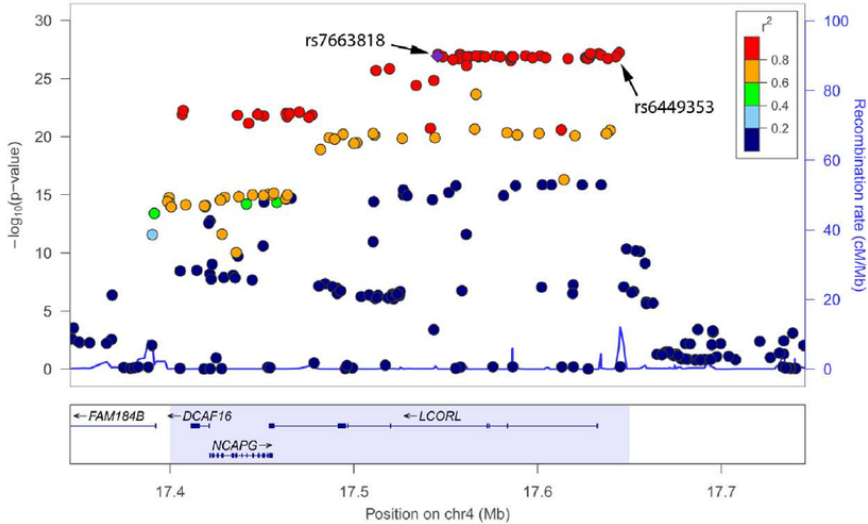



**Figure:** N'Diaye *et al*. (2011) PLoS Genet.

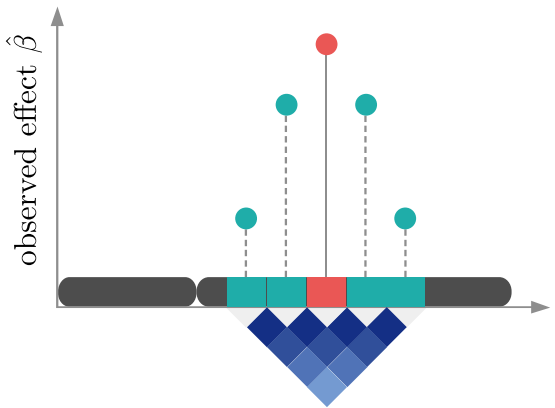
# Objectives

Statistical fine-mapping **aids in** the identification of causal variants, in order to

- interpret association signals (pinpoint to genes)
- understand biological function of a variant
- elucidate genetic architecture of complex and molecular phenotypes

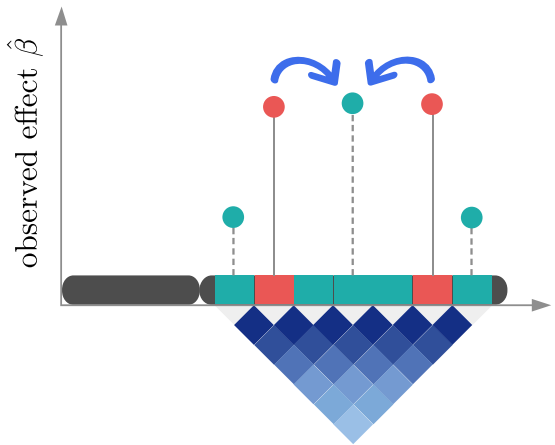
# Identify non-zero effect (“causal”) variables

Simply pick the **top** association in an LD block? Maybe?



# Identify non-zero effect (“causal”) variables

Simply pick the **top** association in an LD block? ... or not!



# Architecture: sparse effects, polygenic background

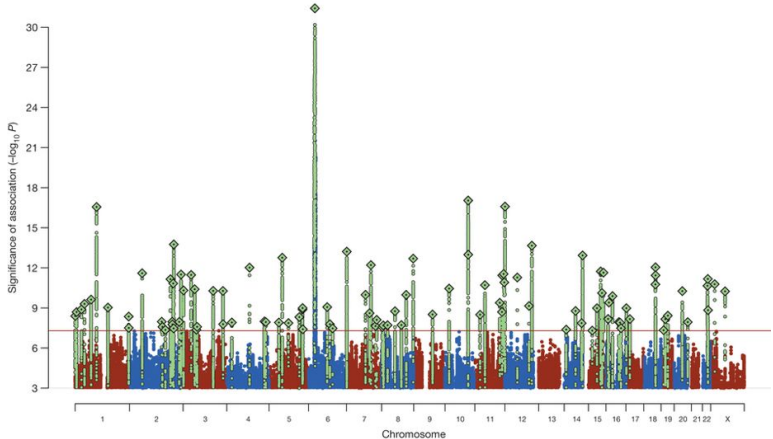



**Figure:** O'Donovan *et al*. (2014) Nature

# Challenge: large biobank sample applications

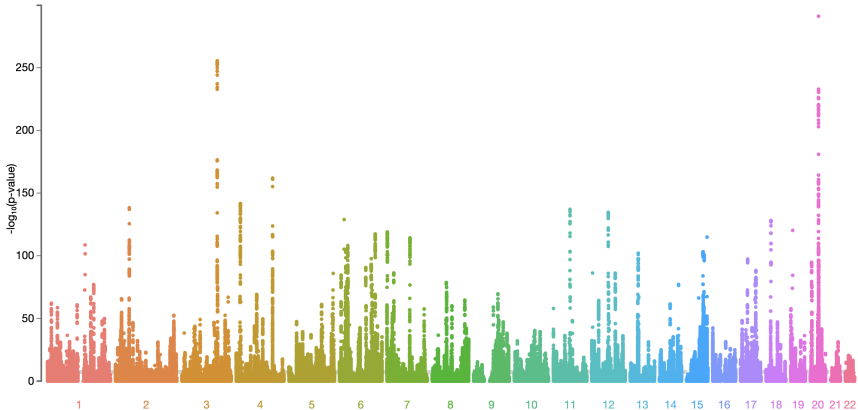



**Figure:** UK Biobank height GWAS, `http://nealelab.is/uk-biobank`

# Toy example, concepts and conditional analysis

# Single Causal variant model

Effect variable (red) correlated with non-effect variable (blue)

| | Case | | Control | |
|---|---|---|---|---|
| SNP | G1 | G2 | G1 | G2 |
| rs1 | 1200 | 800 | 1000 | 1000 |
| rs2 | 1190 | 809 | 1000 | 1000 |

# Single Causal variant model

Compute likelihood ratios (LR) $H_1$ vs $H_0$,

| | Case | | Control | | | |
|---|---|---|---|---|---|---|
| SNP | G1 | G2 | G1 | G2 | LR | p-value |
| rs1 | 1200 | 800 | 1000 | 1000 | $6.15 \times 10^{8}$ | $1.99 \times 10^{-10}$ |
| rs2 | 1190 | 809 | 1000 | 1000 | $0.94 \times 10^{8}$ | $1.37 \times 10^{-9}$ |

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| rs2 | 1190 | 809 | 1000 | 1000 | $0.94 \times 10^8$ | $1.37 \times 10^{-9}$ |

Probability of association assuming **one effect variable**,

$$\frac{\text{LR}_1}{\text{LR}_1 + \text{LR}_2} = \frac{6.15}{6.15 + 0.94} = 0.87 \quad \frac{\text{LR}_2}{\text{LR}_1 + \text{LR}_2} = \frac{0.94}{6.15 + 0.94} = 0.13$$

# Per variable contingency table analysis, likelihood ratio test

For a $2 \times 2$ table with observed frequencies $O_{ij}$ and expected frequencies $E_{ij}$:

$$G = 2 \sum_{i,j} O_{ij} \log \left( \frac{O_{ij}}{E_{ij}} \right)$$

Under $H_0$ (independence):

$$E_{ij} = \frac{(\text{row total}_i)(\text{column total}_j)}{n}$$

Likelihood Ratio:

$$LR = \exp(G/2)$$

The test statistic G follows a $\chi^2$ distribution with 1 degree of freedom under $H_0$

# Per variable contingency table analysis, R code

```r
# returns likelihood ratio and p-value of H_1 vs H_0
get_2x2_lrt = function(tbl) {
    tbl = as.table(matrix(tbl, 2,2,
      dimnames=list(status=c('case','control'),
      genotype=c('minor_allele','major_allele'))))
    test = MASS::loglm(~status+genotype,data=tbl)
    lr = exp(test$lrt / 2)
    pval = 1 - pchisq(test$lrt, df=1)
    return(list(lr=lr, pval=pval))
}
res1 = get_2x2_lrt(c(1200,800,1000,1000))
res2 = get_2x2_lrt(c(1190,809,1000,1000))
```

# Single Causal variant model Bayesian variable selection

Compute Bayes factors (BF) $H_1$ vs $H_0$,

| | Case | | Control | | | |
|---|---|---|---|---|---|---|
| SNP | G1 | G2 | G1 | G2 | BF | $\log_{10}$BF |
| rs1 | 1200 | 800 | 1000 | 1000 | $2.40 \times 10^7$ | 7.38 |
| rs2 | 1190 | 809 | 1000 | 1000 | $0.36 \times 10^7$ | 6.56 |

# Single Causal variant model Bayesian variable selection

| | Case | | Control | | | |
|---|---|---|---|---|---|---|
| SNP | G1 | G2 | G1 | G2 | BF | $\log_{10}$BF |
| rs1 | 1200 | 800 | 1000 | 1000 | $2.40 \times 10^7$ | 7.38 |
| rs2 | 1190 | 809 | 1000 | 1000 | $0.36 \times 10^7$ | 6.56 |

**Posterior inclusion probability** of association assuming **one effect variable**,

$$\text{PIP}_1 = \frac{\text{BF}_1}{\text{BF}_1 + \text{BF}_2} = \frac{2.40}{2.40 + 0.36} = 0.87 \quad \text{PIP}_2 = \frac{\text{BF}_2}{\text{BF}_1 + \text{BF}_2} = \frac{0.36}{2.40 + 0.36} = 0.13$$

# Bayesian analysis for $2 \times 2$ tables, model specification

Data: $\mathbf{n} = (n_{11}, n_{12}, n_{21}, n_{22})$

Multinomial Likelihood:

$$\mathbf{n} \sim \text{Multinomial}(N, \boldsymbol{\theta})$$

$$P(\mathbf{n}|\boldsymbol{\theta}) = \frac{N!}{\prod_{i,j} n_{ij}!} \prod_{i,j} \theta_{ij}^{n_{ij}}$$

Independence Model ($H_0$):

$\theta_{ij} = \pi_i \times \gamma_j$

(Requires row-column independence)

$\boldsymbol{\pi} \sim \text{Dirichlet}(a, a); \boldsymbol{\gamma} \sim \text{Dirichlet}(a, a)$

Association Model ($H_1$):

$\boldsymbol{\theta} \sim \text{Dirichlet}(a, a, a, a)$

(Allows for all possible associations)

Bayes Factor:

$$\text{BF} = \frac{P(\text{data}|H_1)}{P(\text{data}|H_0)}$$

$$= \frac{\int P(\mathbf{n}|\boldsymbol{\theta}) p(\boldsymbol{\theta}|H_1) d\boldsymbol{\theta}}{\int P(\mathbf{n}|\boldsymbol{\theta}) p(\boldsymbol{\theta}|H_0) d\boldsymbol{\theta}}$$

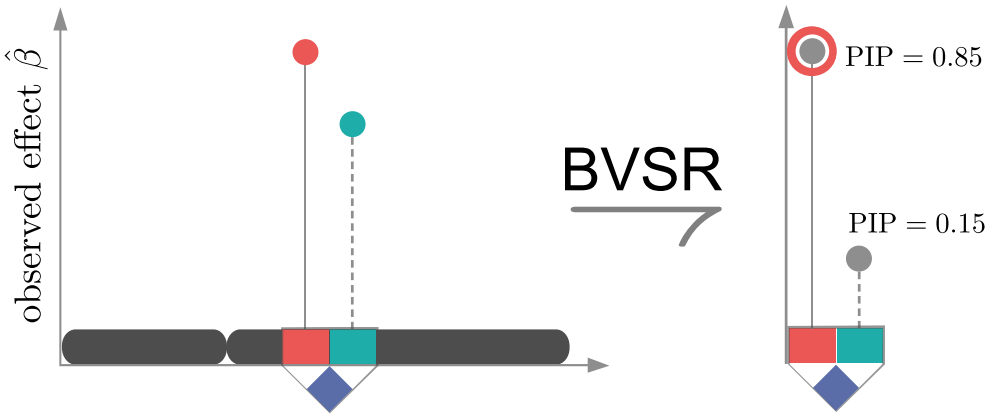
Common choice: $a = 1$ (uniform prior)

# Bayesian contingency table analysis, R code

```r
# returns Bayes factor of H_1 vs H_0
get_2x2_bf = function(tbl, prior_a = 1) {
    n = matrix(tbl, 2, 2, byrow=T)
    dimnames(n) = list(status=c('case','control'),
                       genotype=c('minor','major'))
    res = BayesFactor::contingencyTableBF(n, sampleType="indepMulti",
          fixedMargin="rows", priorConcentration=prior_a)
    bf = exp(res@bayesFactor$bf)
    return(list(bf = bf, log10bf = log(bf)/log(10)))
}
res1 = get_2x2_bf(c(1200,800,1000,1000))
res2 = get_2x2_bf(c(1190,809,1000,1000))
```

# Bayesian variable selection: PIP

Computes **Posterior Inclusion Probability** (PIP)



# Bayesian variable selection: PIP

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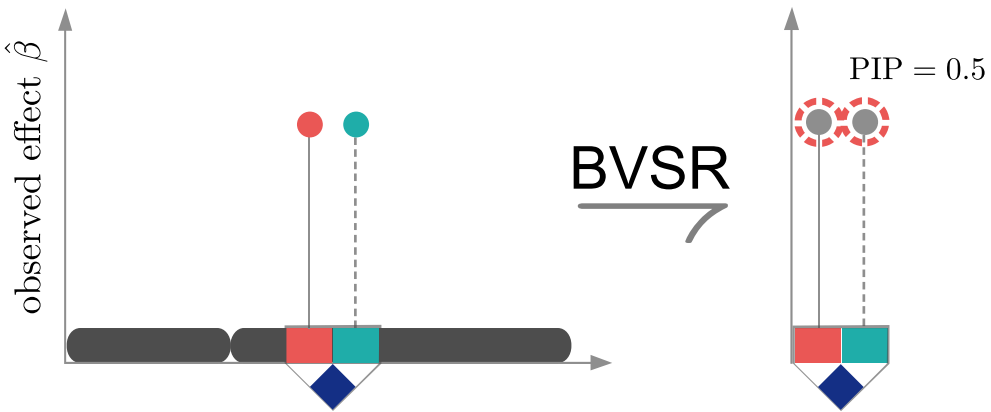
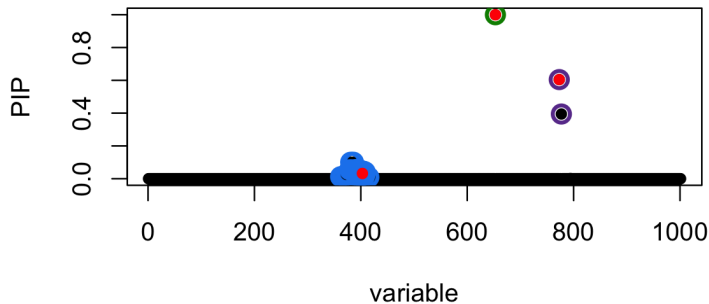


# Bayesian variable selection: Credible Sets

“Clusters” of signals to account for correlations between variables (eg LD)



# Bayesian variable selection: Credible Sets

- **95% credible set** $S$: $\Pr(\textit{effect variable in } S) \geq 95\%$
- *e.g.* , "Single effect" model:

$$\sum_{j \in S} PIP_{(j)} \geq 95\%$$

  where $PIP_{(j)}$'s are in descending order.
- Formal definition: Wang *et al.* (2020) J. R. Stat. Soc. B

# Multiple causal effects: step-wise search

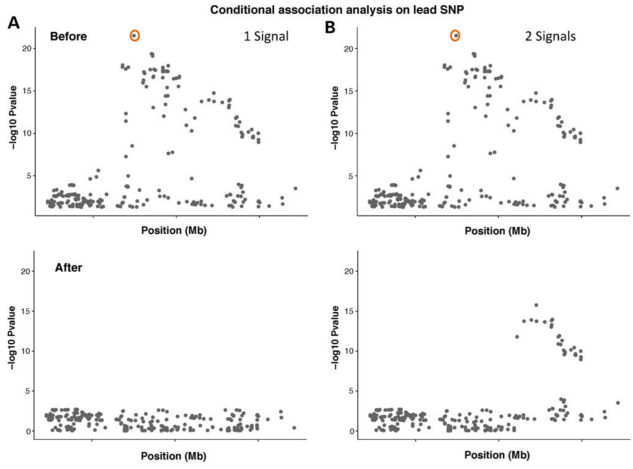



**Figure:** Spain and Barrett (2015) Hum. Mol. Genet.

# A simple frequentist conditional analysis

**Forward selection algorithm**

1. For each SNP fit a simple linear regression model
2. Select the SNP $j$ that has the largest model likelihood
3. Form residuals $\boldsymbol{y}' := \boldsymbol{y} - \boldsymbol{X}_j \hat{b}_j$, and repeat

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**A greedy algorithm to choose the "best" SNPs, but is incapable of capturing multiple SNPs in LD**

# (Approximate) COnditional and JOint Analysis

**Method overview**

- Frequentist framework for summary statistics
- Implements stepwise variable selection
- GCTA software implementation (Yang et al. 2011)

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**Method overview**

- Frequentist framework for summary statistics
- Implements stepwise variable selection
- GCTA software implementation (Yang et al. 2011)

**Limitation**

- May “over-regress” on lead variants when multiple causal variants exist in partial LD
- Example: if two SNPs in partial LD ($r^2 = 0.3$) are both causal:
  - First SNP captures most association
  - Second SNP effect underestimated after conditioning
- Result: Potentially missing independent functional variants

*Need to sample from a distribution, not selecting the best signal*

# COJO algorithm, Yang et al. (2012) Nat. Genet.

1. Start with a model with the most significant SNP in the single-SNP meta-analysis across the whole genome with *P* value below a cutoff *P* value, such as $5 \times 10^{-8}$.
2. For the *i*th step, calculate the *P* values of all the remaining SNPs conditional on the SNP(s) that have already been selected in the model. To avoid problems due to collinearity, if the squared multiple correlation between a SNP to be tested and the selected SNP(s) is larger than a cutoff value, such as 0.9, the conditional *P* value for that SNP will be set to 1.
3. Select the SNP with minimum conditional *P* value that is lower than the cutoff *P* value. However, if adding the new SNP causes new collinearity problems between any of the selected SNPs and the others, we drop the new SNP and repeat this process.
4. Fit all the selected SNPs jointly in a model and drop the SNP with the largest *P* value that is greater than the cutoff *P* value.
5. Repeat (2), (3) and (4) until no SNPs can be added or removed from the model.

# Quantify uncertainty: which to select among many correlated variables

**Bayesian forward selection algorithm**

1. For each SNP $j$, fit a simple Bayesian linear regression model to get Bayes Factor $\text{BF}_j$
2. Form weight for each SNP, $w_j \propto \text{BF}_j$
3. Form residuals $\boldsymbol{y}' := \boldsymbol{y} - \sum_j w_j \boldsymbol{X}_j \hat{b}_j$, and repeat

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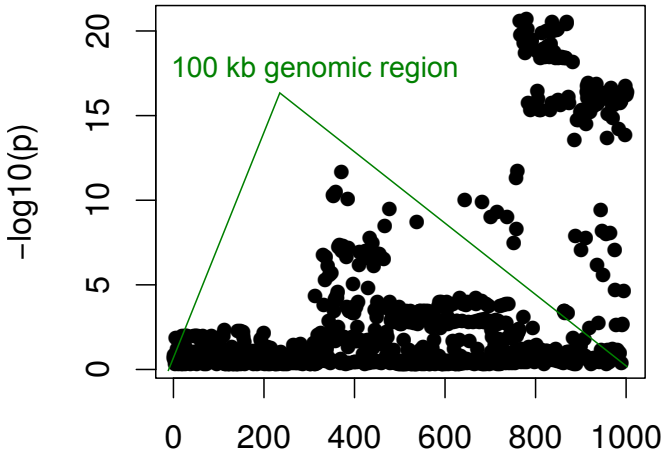
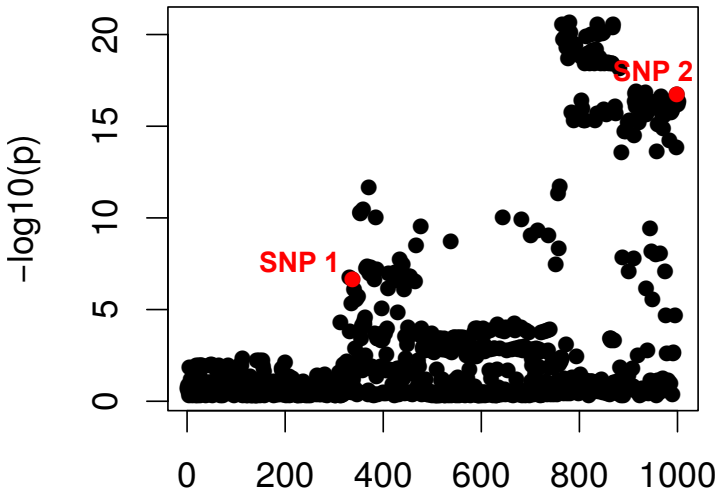
**Still the same limitation: what if a "bad decision" was made early on?**

# A motivating example

Data available as `data(susieR::N2finemapping)`



# A motivating example

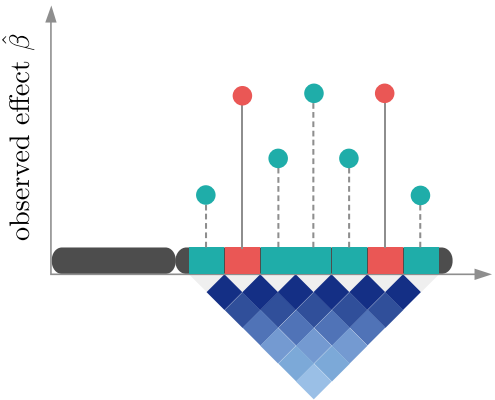


# Fine-mapping using individual level data

Bayesian Variable Selection Regression and the SUm of SIngle Effects model

# Detecting multiple effect variables

**Intuition**: A model involving the two effect variables should fit the data better than that involving only the top variable.



| Software | Trait type[a] | Input covariates[b] | Uses summary statistics? | Maximum number of causal variants[c] | Input annotation? | Causal search | Main output |
|---|---|---|---|---|---|---|---|
| BIMBAM v1.0 | qt and binary | No | No | Fixed | No | Exhaustive | Bayes factor |
| mvBIMBAM v1.0.0 | mqt | No | Yes | 1 | No | Exhaustive | Bayes factor |
| SNPTEST v2.5.4-beta3 | qt, binary, mqt and multinomial | No | No | 1 | No | Exhaustive | Bayes factor |
| piMASS v0.9 | qt and binary | No | No | Computed | No | MCMC | Bayes factor and PIP |
| BVS v4.12.1 | Binary | Yes | No | Computed | Yes | MCMC | Bayes factor and PIP |
| FM-QTL | qt | No | No | Computed | Yes | MCMC | Bayes factor and PIP |
| DAP v1.0.0 | qt | Yes | Yes | 1, fixed and computed | Yes | Exhaustive | Bayes factor and PIP |
| Fine-mapping | Multinomial | Yes | No | Computed | No | Greedy | PIP |
| Trinculo | Multinomial | Yes | No | Computed | No | Greedy | Bayes factor and PIP |
| BayesFM | Binary | Yes | No | 20 | No | MCMC | PIP |
| ABF | qt and binary[d] | Yes | Yes | 1 | No | Exhaustive | Bayes factor |
| fgwas v0.3.6 | qt and binary[d] | No | Yes | 1 | Yes | Exhaustive | Bayes factor and PIP |
| CAVIAR/eCAVIAR | qt and binary[d] | No | Yes | Fixed | No | Exhaustive | ρ probability confidence set and PIP |
| PAINTOR v3.0 | qt, binary[d] and mqt | No | Yes | Fixed and computed | Yes | Exhaustive and MCMC | Bayes factor and PIP |
| CAVIARBF v0.2.1 | qt and binary[d] | No | Yes | Fixed | Yes | Exhaustive | Bayes factor and PIP |
| FINEMAP v1.1 | qt and binary[d] | No | Yes | Fixed | No | Shotgun stochastic search | Bayes factor and PIP |
| JAM in R2BGLiMS v0.1 | qt and binary[d] | No | Yes | Fixed and computed | No | Exhaustive and MCMC | Bayes factor and PIP |

**Figure:** Schaid *et al*. (2018) Nat. Rev. Genet.

# Bayesian Variable Selection Regression (BVSR) model

Fine-mapping is a particular multiple regression problem:

$$\boldsymbol{y}_{n\times 1} = \boldsymbol{X}_{n\times p}\boldsymbol{b}_{p\times 1} + \boldsymbol{e}_{n\times 1}$$

- $\boldsymbol{b}$ is sparse: most of its elements are 0's
- Columns of $\boldsymbol{X}$ are very correlated

# BVSR posterior

Assess **combinations** of variables

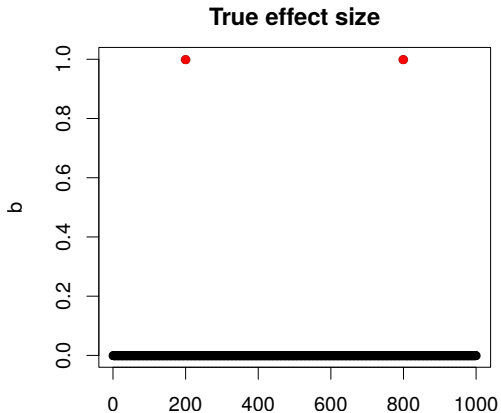
| | SNPs | 1 | 2 | 3 | 4 | 5 | $\cdots$ | Probability |
|---|---|---|---|---|---|---|---|---|
| model configurations | | 1 | 0 | 1 | 0 | 0 | $\cdots$ | 0.25 |
| | | 1 | 0 | 0 | 1 | 0 | $\cdots$ | 0.25 |
| | | 0 | 1 | 1 | 0 | 0 | $\cdots$ | 0.25 |
| | | 0 | 1 | 0 | 1 | 0 | $\cdots$ | 0.25 |

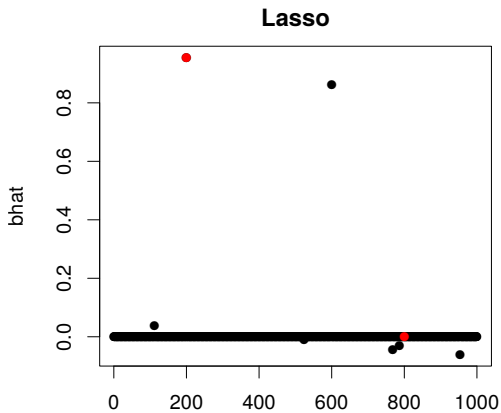
- $\text{PIP}_j := Pr(z_j \text{ is non-zero})$

$$\text{PIP} = (0.5, 0.5, 0.5, 0.5, 0, \cdots)$$

# BVSR quantifies uncertainty in variable selection

$b_1 \neq 0$ or $b_2 \neq 0$, and $b_3 \neq 0$ or $b_4 \neq 0$.



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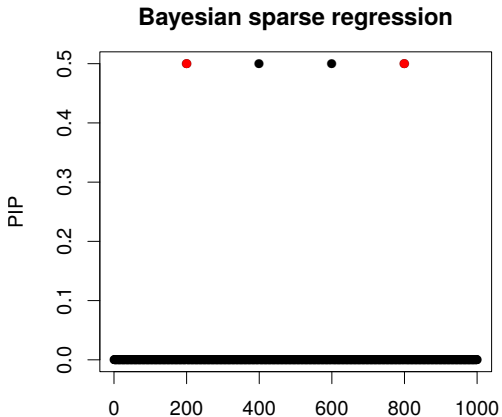
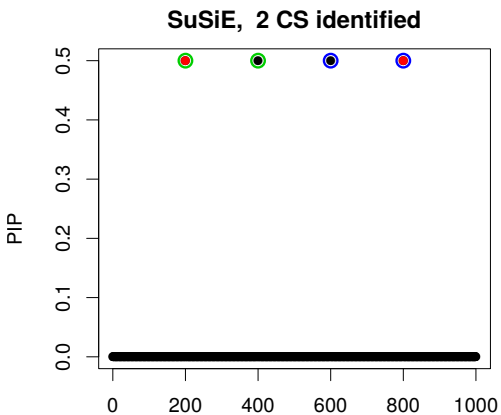
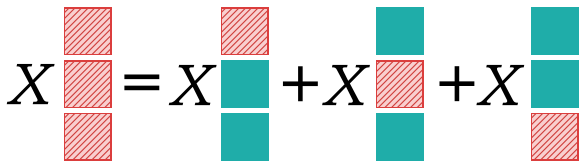


# The Sum of Single Effects model (*SuSiE*)

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{b} + \boldsymbol{e}$$

$$\boldsymbol{b} = \sum_{l=1}^{L} \boldsymbol{b}_l$$



Wang *et al*. (2020) J. R. Stat. Soc. B

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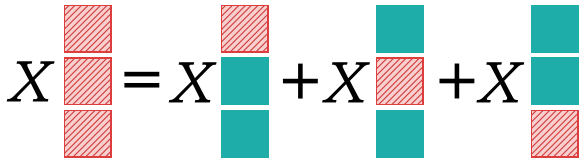
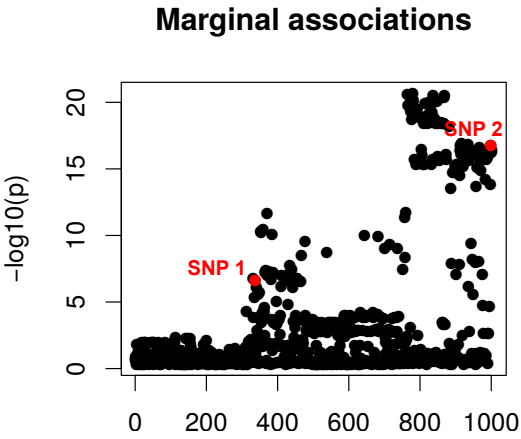


**A variational approximation to posterior under *SuSiE***

$$q(\boldsymbol{b}_1, \ldots, \boldsymbol{b}_L) = \prod_l q_l(\boldsymbol{b}_l)$$

- $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_L$ are treated as **independent** *a posteriori*.
- **Do not** assume $q_l$ factorizes over the elements of $\boldsymbol{b}_l$.

# Real-world example illustrated



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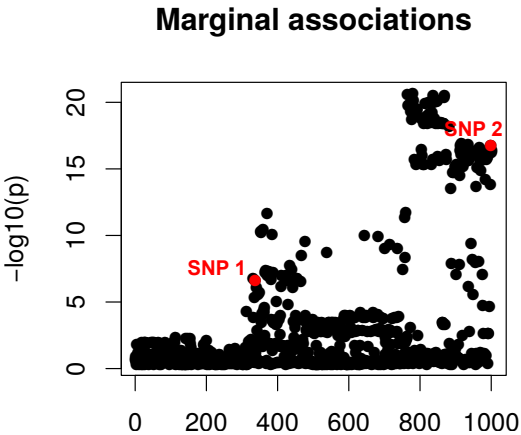


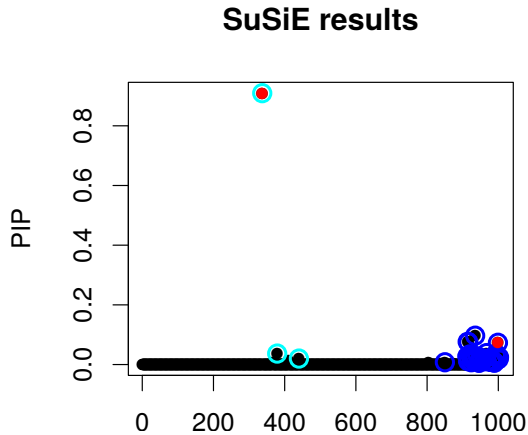

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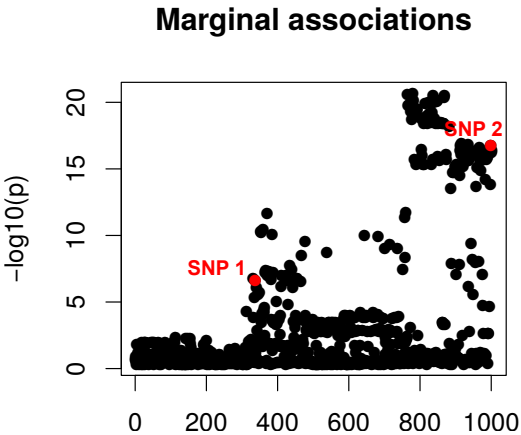


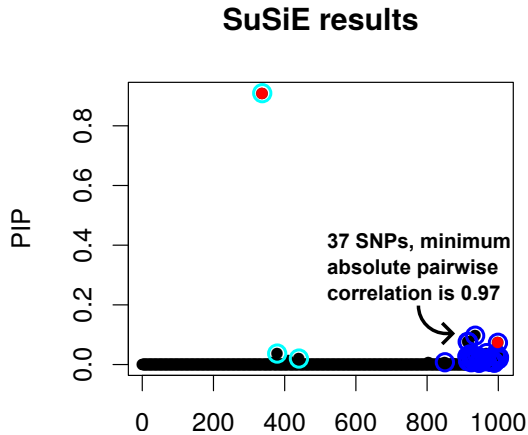

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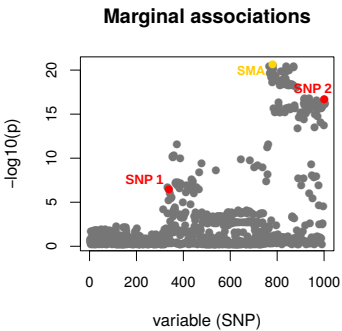
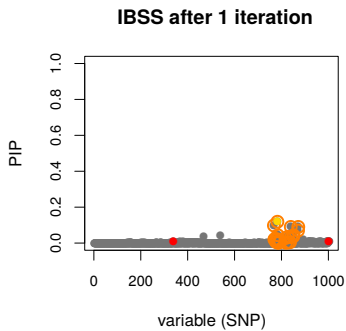
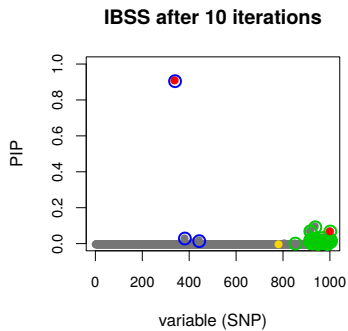
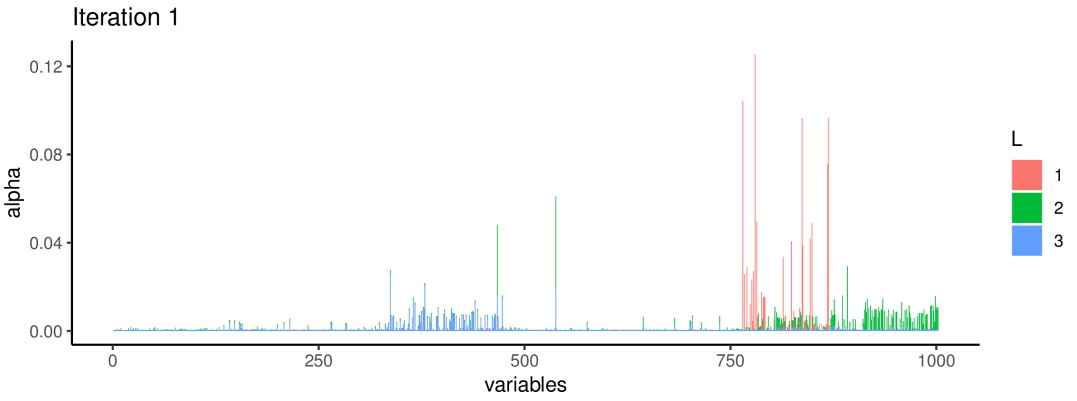


# The IBSS algorithm iterations breakdown



# Fine-mapping using association summary statistics

with practical considerations in real-world data

# Summary statistics in association studies

- $X$, genotype matrix
- $Y$, phenotype matrix, can be multiple traits
- $X^TY$, association results — effect size estimate
- $X^TX$, LD matrix
- $XX^T$, genomic relatedness matrix, reflects kinship
- $Y^TY$, trait correlation, relevant in multi-trait analysis and integration

# Reasons to work with summary statistics

Advantage over full data (genotypes and phenotypes):

- Easier to obtain and share with others
- Convenient to use: QC and data wrestling barely needed
- Computationally suitable for large-sample problems
  - $\mathcal{O}(p^2)$ (summary statistics) $\ll \mathcal{O}(np)$ (full data)
  - when sample size $n \gg$ variants in fine-mapped region $p$

Suggested reading: Pasaniuc and Price (2017) Nat. Rev. Genet.

# Association testing marginal effects summary statistics

$z$-scores from univariate association studies:

$$\hat{z}_j := \hat{\beta}_j / s_j,$$

where

$$\hat{\beta}_j := (\boldsymbol{x}_j^\mathsf{T}\boldsymbol{x})^{-1}\boldsymbol{x}^\mathsf{T}\boldsymbol{y} \quad s_j := \sqrt{\hat{\sigma}_j^2 (\boldsymbol{x}_j^\mathsf{T}\boldsymbol{x})^{-1}}$$

- **Sufficient** statistics: $\boldsymbol{x}^\mathsf{T}\boldsymbol{x}$, $\boldsymbol{x}^\mathsf{T}\boldsymbol{y}$, $\hat{\sigma}_j^2$
- $\hat{\sigma}_j^2$ is the estimated residual variance from regressing $\boldsymbol{y}$ on $\boldsymbol{x}_j$

# Approximate multiple linear regression from summary statistics

**Ordinary least-squares:**

$$\beta_{\text{joint}} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{y}) = (N^{-1}\mathbf{X}'\mathbf{X})^{-1}(N^{-1}\mathbf{X}'\mathbf{y}) = (\mathbf{R}_{\text{GWAS}})^{-1}\mathbf{D}^{-1}\beta_{\text{GWAS}}$$

where:

- $\mathbf{X}$ is the matrix a $N \times M$ matrix of genotypes (at the focal locus) in the GWAS sample
- $\mathbf{y}$ is the vector of phenotypes for the $N$ individuals in the GWAS
- $\mathbf{R}_{\text{GWAS}}$ is the correlation matrix between SNPs in the GWAS sample
- $\beta_{\text{GWAS}}$ is the vector marginal SNP effects
- $\mathbf{D}$ is a diagonal matrix: $D_j = \text{var}(X_j)$ is the variance of allele counts at SNP $j$ in the GWAS sample, as a function of sample size and MAF

# Approximate multiple linear regression from summary statistics

**Ordinary least-squares:**

$$\boldsymbol{\beta}_{\text{joint}} = (\mathbf{R}_{\text{GWAS}})^{-1}\mathbf{D}_{\text{GWAS}}^{-1}\boldsymbol{\beta}_{\text{GWAS}}$$

**Issue:** However, $\mathbf{R}_{\text{GWAS}}$ and $\mathbf{D}_{\text{GWAS}}$ may not always been available.

**Solutions:** approximate $\mathbf{R}_{\text{GWAS}}$ and $\mathbf{D}_{\text{GWAS}}$ from a sample of individuals with same genetic ancestries as the GWAS sample.

$$\boldsymbol{\beta}_{\text{joint}} \approx (\mathbf{R}_{\text{REF}})^{-1}\mathbf{D}_{\text{REF}}^{-1}\boldsymbol{\beta}_{\text{GWAS}}$$

Crucial assumption in many summary statistics-based methods.

# Regression model for z-scores

$$\hat{\boldsymbol{z}} \sim N(\hat{\boldsymbol{R}}\boldsymbol{z}, \hat{\boldsymbol{R}})$$

Assumptions:

1. Heritability of any single SNP is small
2. $\hat{\boldsymbol{R}}$ is sample genotypic correlation for **the same study**
3. Genotypes used to computed $\boldsymbol{z}$ and $\hat{\boldsymbol{R}}$ are accurate

# Regression for with $\hat{\beta}$ and SE$(\hat{\beta})$

The $\hat{z}$ model (exchangable z-scores):

$$\hat{\boldsymbol{z}} \sim N(\hat{\boldsymbol{R}}\boldsymbol{z}, \hat{\boldsymbol{R}})$$

The $\hat{b}, \hat{s}$ model (exchangable effects):

$$\hat{\boldsymbol{b}}|\hat{\mathbf{s}} \sim N(\hat{\mathbf{S}}\hat{\boldsymbol{R}}\hat{\mathbf{S}}^{-1}\boldsymbol{b}, \hat{\mathbf{S}}\hat{\boldsymbol{R}}\hat{\mathbf{S}})$$

- $\hat{z}$ model: lower MAF variants have larger effects
- $\hat{b}, \hat{s}$ model: effect sizes are the same regardless of MAF
- $\hat{b}, \hat{s}$ model takes sample size into consideration
  - No longer have to assume small effect per SNP

Suggested reading: Zhu and Stephens (2017) AoAS

# Properties of per SNP $z$ scores

- $z$-score for a SNP depends on effects of both itself and other correlated SNPs:

$$\mathrm{E}(\hat{z}_j|\hat{\boldsymbol{R}}) = \sum_{i=1}^{p} r_{ij} z_j.$$

**GWAS marginal effects are biased due to LD!**

- $z$-scores are correlated,

$$\mathrm{Cor}(\hat{z}_j, \hat{z}_k) = r_{jk}, \forall j, k$$

- Application: LD score regression (LDSC)

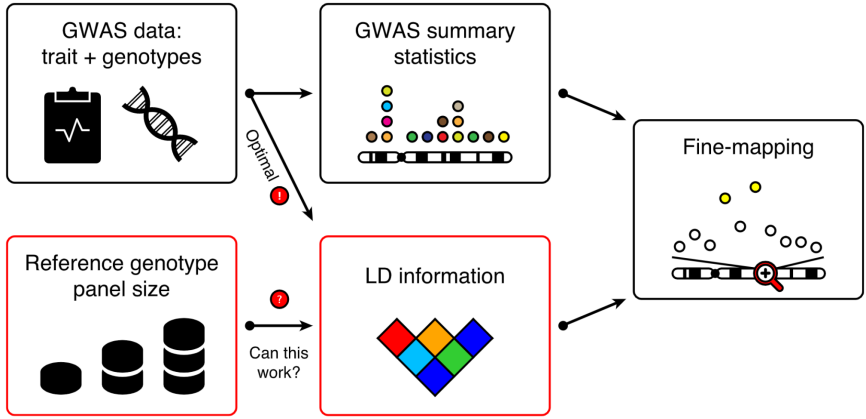



**Figure:** Benner *et al*. (2017) Am. J. Hum. Genet.

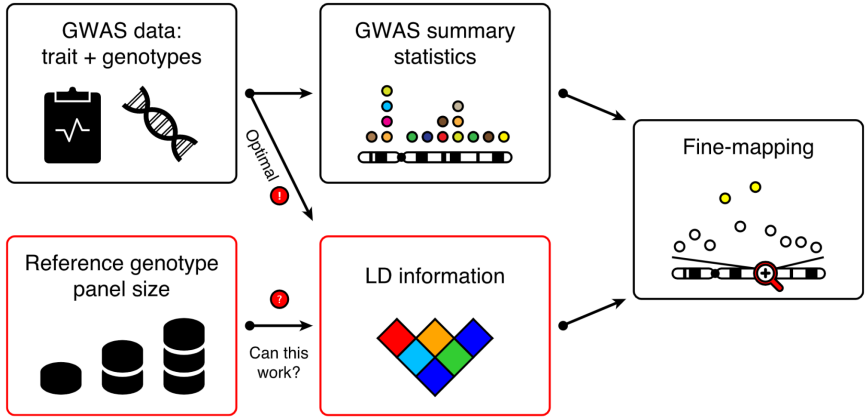



**Figure:** Benner *et al.* (2017) Am. J. Hum. Genet.

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

$$f(\boldsymbol{\beta}|\text{Summary data}) \propto f(\text{Summary data}|\boldsymbol{\beta}) \times f(\boldsymbol{\beta})$$

# Single Causal variant model fine-mapping using summary statistics

- $m$ SNPs in the region to fine-map
- Prior = each SNP has the same probability to be causal
- Posterior:

$$P(C_j|Z_1, ..., Z_m) = \frac{\exp(\frac{Z_j^2}{2})}{\sum_{k=1}^{m} \exp(\frac{Z_k^2}{2})}$$

# *SuSiE* Regression with Summary Statistics (RSS)

"Single effects": $\boldsymbol{z}_l$'s

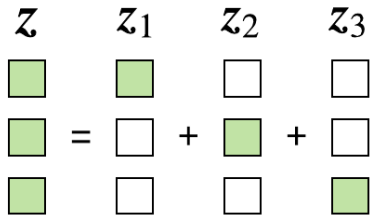
$$\hat{\boldsymbol{z}} \sim N(\hat{\boldsymbol{R}}\boldsymbol{z}, \hat{\boldsymbol{R}})$$

$$\boldsymbol{z} = \sum_{l=1}^{L} \boldsymbol{z}_l$$

$$\boldsymbol{z}_l = \gamma_l z_l$$

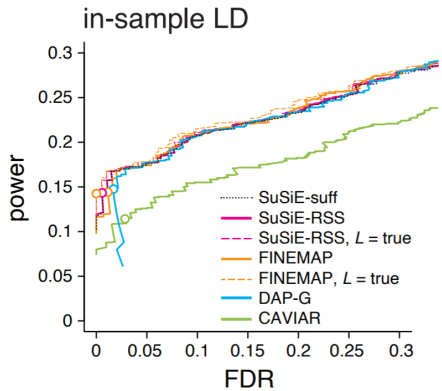
$$z_l \sim N(0, \omega_l^2)$$

$$\gamma_l \sim \text{Mult}(1, \boldsymbol{\pi})$$




Suggested reading: Zou et al (2022) PLoS Genet.

# Summary statistics fine-mapping methods comparison



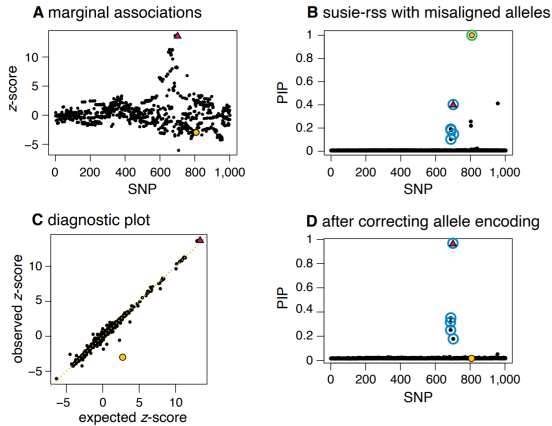

**In practice people often use SuSiE RSS, Zou *et al*. (2022) PLoS Genet + FINEMAP, Benner *et al*. (2016) Bioinformatics**

# Impact of bioinformatics error: allele flips

What is allele flip?

- Different allele encoding between GWAS and LD reference
- *e.g*. AA=0, AC=1, CC=2 in GWAS; AA=2, AC=1, CC=0 in LD reference genotype
- A challenging problem coupled with strand flip, when merging sequence data from different platforms

# Impact of allele flips



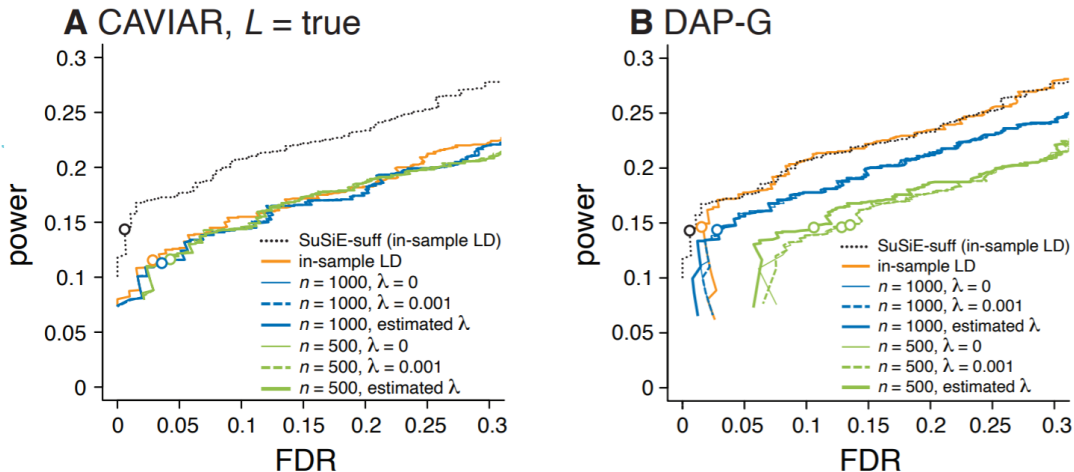

Zou *et al*. (2022) PLoS Genet.

# Addressing the allele flip challenge

R packages,

- `susieR::susie_rss()` function implements a diagnosis
- `bigsnpr::snp_match()` function implements a basic allele matching for two sets of summary statistics
- `MungeSumstats` R package provides a suite of tools to standardize association testing summary statistics.

# Impact of mis-matched LD reference: PIP



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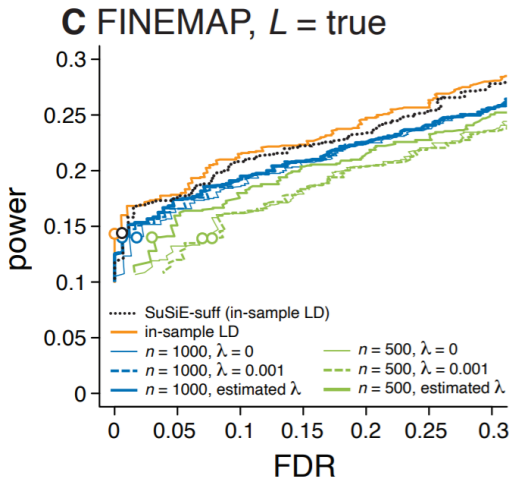
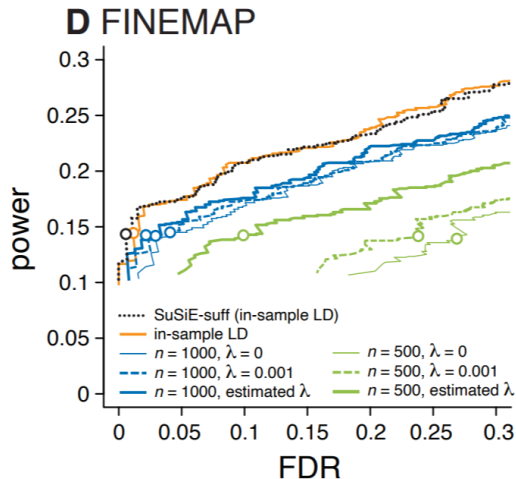
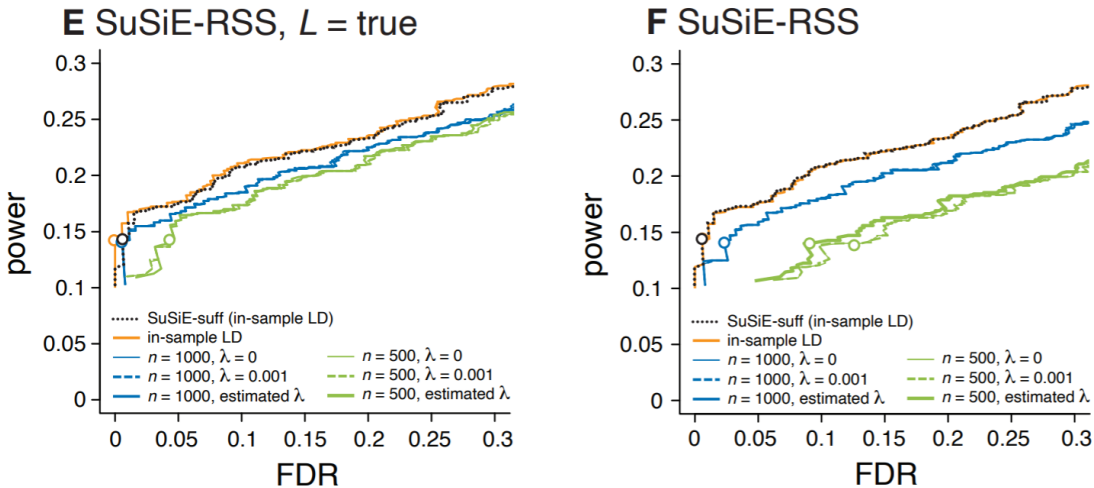
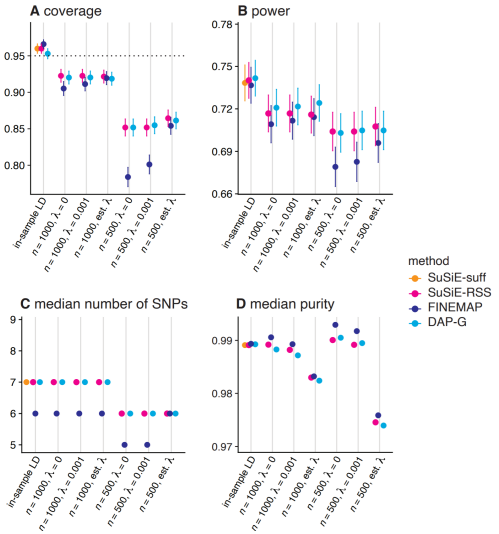


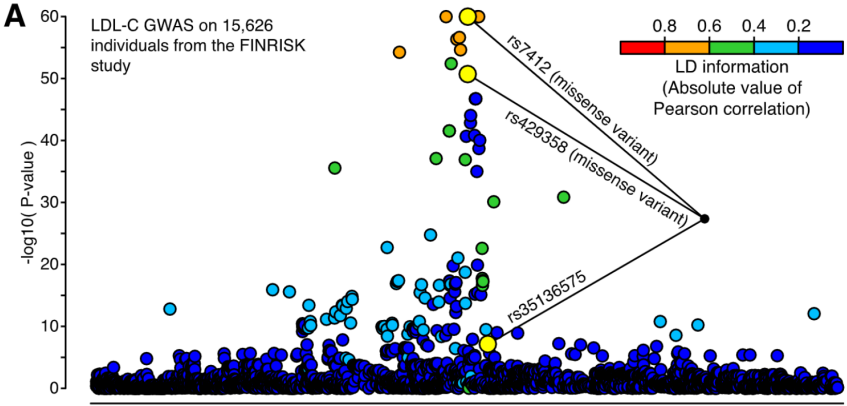
# Impact of mis-matched LD reference: PIP



# Impact of mis-matched LD reference: credible sets

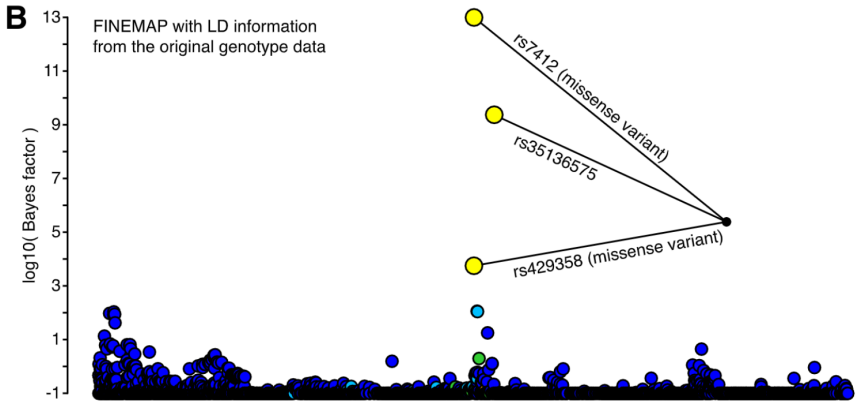


# Impact of mis-matched LD reference: real data



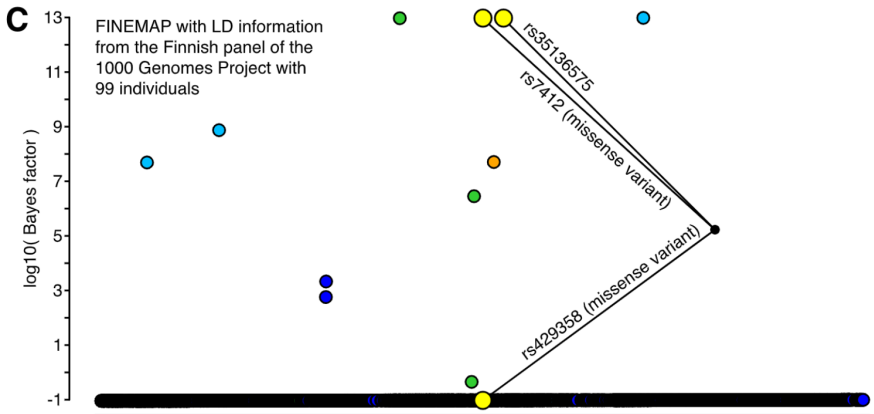

Benner *et al*. (2017) Am. J. Hum. Genet.

# Impact of mis-matched LD reference: real data




Benner *et al*. (2017) Am. J. Hum. Genet.

# Impact of mis-matched LD reference: real data




Benner *et al*. (2017) Am. J. Hum. Genet.

# Fine-mapping in meta-analysis: overview

**GWAS cohorts**

Cohort 1

Cohort 2

⋮

Cohort N

Across constituent cohorts, inter-cohort heterogeneity could arise from:

- Genuine biological mechanisms
  - Population-specific variants
  - GxG and GxE interactions
- Phenotyping
  - Different diagnosis criteria
  - Different proportion of subtypes
  - Different measurement protocols
- Genotyping and imputation
  - Different genotyping array
  - Different imputation reference panel
  - Different imputation quality
- Quality control (QC)
  - Different thresholds for MAF, imputation quality, etc.
- GWAS
  - Different statistical model and software

**Meta-analysis**

Effect models:
- Fixed-effect
- Random-effect

Ancestries:
- Single-ancestry
- Multi-ancestry

Typically, both pre- and post-meta-analysis QC are applied to summary statistics (**Data S1**).

For each locus

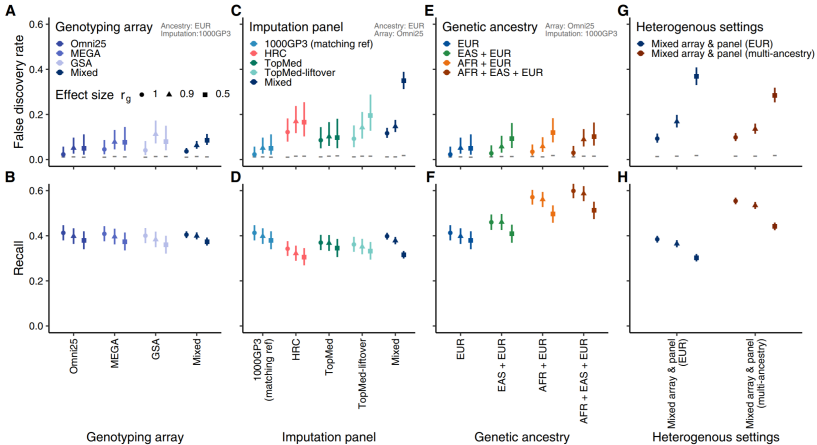
**Fine-mapping**

Summary statistics-based methods include:
- ABF
- CAVIAR
- PAINTOR
- FINEMAP
- SuSiE

Standard outputs:
- Posterior inclusion probability (PIP)
- 95% credible sets

Additional post-fine-mapping QC is sometimes adopted.

# Key factors of reference data discrepancy in meta-analysis

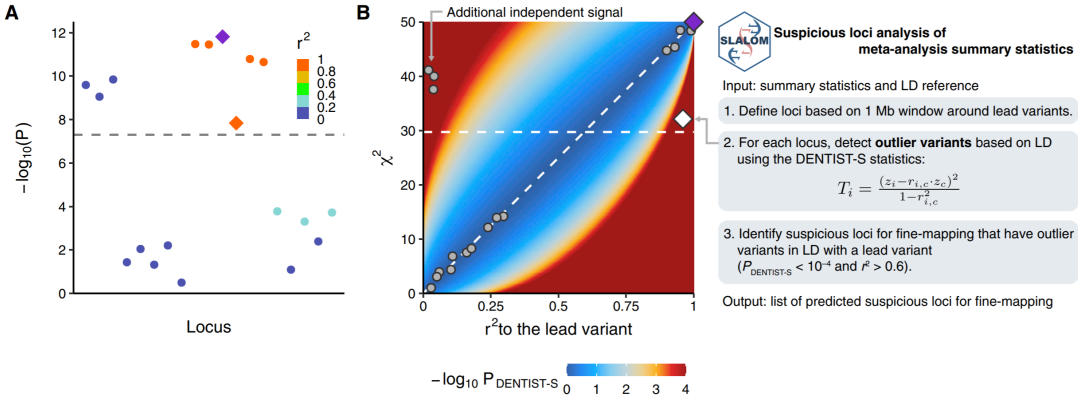



Kanai *et al*. (2022) Cell Genomics

# Fine-mapping in meta-analysis: mismatch diagnosis

1. Compare allele frequencies between LD panel and GWAS, filter out SNPs with too large differences (GCTA: `--diff-freq`)
2. Filter per-SNP sample size outliers
3. Use methods like DENTIST and SLALOM to detect LD inconsistencies

# Fine-mapping in meta-analysis: mismatch diagnosis




Chen *et al*. (2021) Nat. Comm. (DENTIST)

Kanai *et al*. (2022) Cell Genomics (SLALOM)

# Covariate adjustment in LD reference

Consider two GWAS regression analysis:

1. Evaluate SNP effect in “Trait $\sim$ SNP + Age + Sex + PCs”
2. Fit model “Trait $\sim$ Age + Sex + PCs”, compute residual of Trait (remove covariates), and evaluate SNP effect in model Residual_Trait $\sim$ SNP

Are these two analysis equivalent?

# Covariate adjustment in LD reference

Consider two GWAS regression analysis:

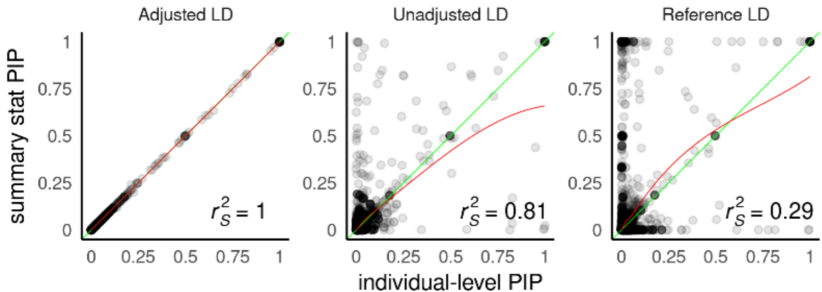
1. Evaluate SNP effect in "Trait $\sim$ SNP + Age + Sex + PCs"
2. Fit model "Trait $\sim$ Age + Sex + PCs", compute residual of Trait (remove covariates), and evaluate SNP effect in model Residual_Trait $\sim$ SNP

They are not equivalent because covariates should also be removed from SNP data: Residual_Trait $\sim$ Residual_SNP

More technical details see McCaw *et al*. (2020) Biometrics

# Kinship (GRM) adjustment in LD reference

Kinship, represented as Genetic Relationship Matrix (GRM) should be removed from genotype data before computing LD reference for fine-mapping with summary statistics




Quick *et al*. (2020) biorxiv

# Functional enrichment in fine-mapped variants

Signals concentrated in tissue / cell specific functional area

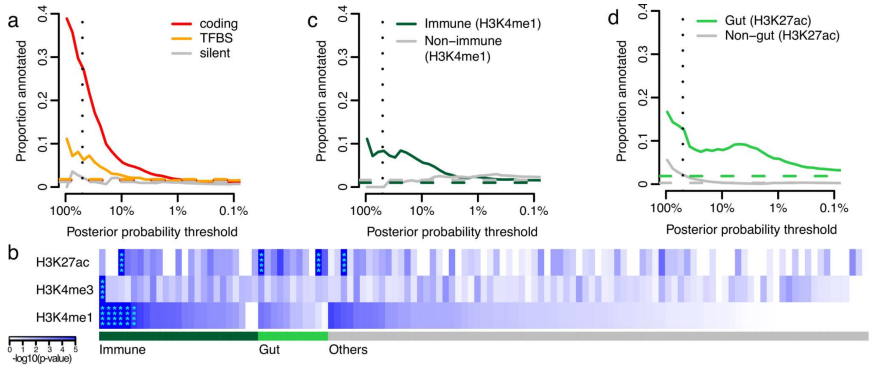
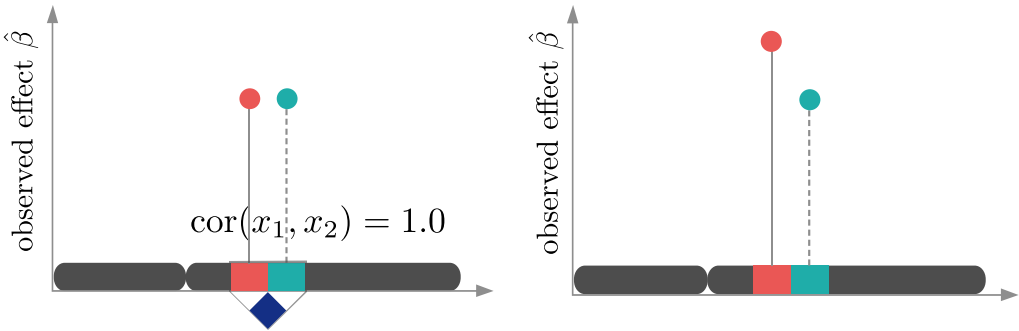



**Figure:** Huang *et al.* (2017) Nature

# Annotations improves fine-mapping resolution




Integrating functional information prioritizes the left SNP.

# Previous example: SNP 1 is causal, SNP 2 in high LD with it

| | Case | | Control | | | |
|---|---|---|---|---|---|---|
| SNP | G1 | G2 | G1 | G2 | BF | $\log_{10}$BF |
| rs1 | 1200 | 800 | 1000 | 1000 | $2.40 \times 10^7$ | 7.38 |
| rs2 | 1190 | 809 | 1000 | 1000 | $0.36 \times 10^7$ | 6.56 |

# When functional annotation is considered

**Posterior inclusion probability** of association assuming **one effect variable**,

$$\text{PIP}_1 = \frac{\text{BF}_1}{\text{BF}_1 + \text{BF}_2} = \frac{2.40}{2.40 + 0.36} = 0.87 \quad \text{PIP}_2 = \frac{\text{BF}_2}{\text{BF}_1 + \text{BF}_2} = \frac{0.36}{2.40 + 0.36} = 0.13$$

What if we determine *a priori* that SNP 1 is **twice as important** as SNP 2?

$$\frac{\mathbf{2} \times \text{BF}_1}{\mathbf{2} \times \text{BF}_1 + \text{BF}_2} = 0.93 \quad \frac{\text{BF}_2}{\mathbf{2} \times \text{BF}_1 + \text{BF}_2} = 0.07$$

# Fine-mapping with functional annotations

Functional annotation in Bayesian Variable Selection Regression:

$$
\begin{aligned}
\boldsymbol{y} &= \boldsymbol{X}\boldsymbol{b} + \boldsymbol{e} \\
\boldsymbol{e} &\sim N(0, \sigma^2 I_n) \\
\gamma_j &\sim \text{Bernoulli}(\boldsymbol{\pi}) \\
\boldsymbol{b}_\gamma | \gamma &\sim g(\cdot) \\
\boldsymbol{b}_{-\gamma} | \gamma &\sim \delta_0
\end{aligned}
$$

Key idea: $\pi$, prior inclusion probability, can be modelled by **enrichment** of functional annotations

# Functionally informed fine-mapping in UK Biobank

In analyses of 49 UK Biobank traits, PolyFun + SuSiE identified >32% more fine-mapped variant–trait pairs compared to using SuSiE alone.

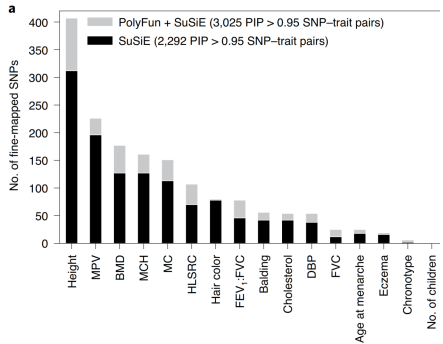
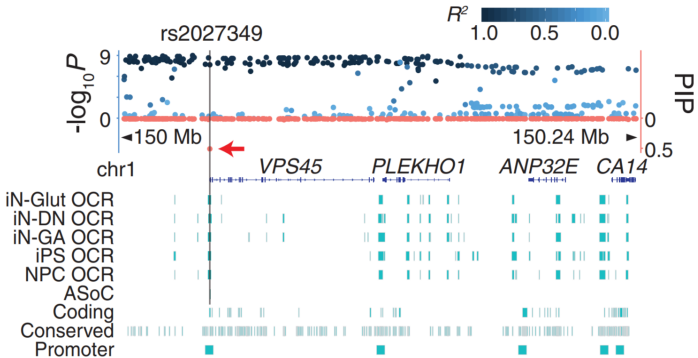



**Figure:** Weissbrod *et al*. (2020) Nat. Genet.

# Example: *SuSiE* with functional informed prior



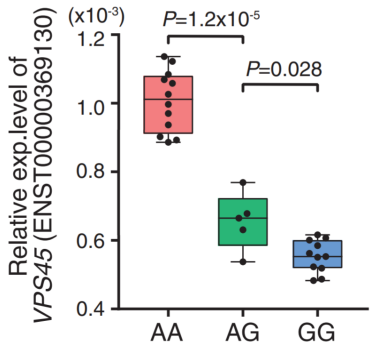




**Figure:** Zhang *et al*. (2020) Science

# Caution: disease specific enrichment

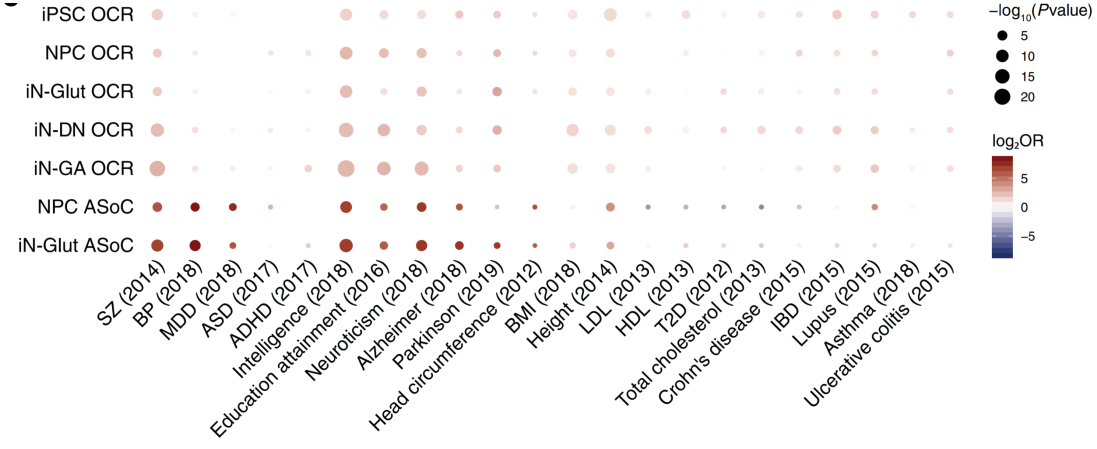



**Figure:** Zhang *et al*. (2020) Science

# Exercise: fine-mapping

# Overview of molecular quantitative trait loci (QTL) studies

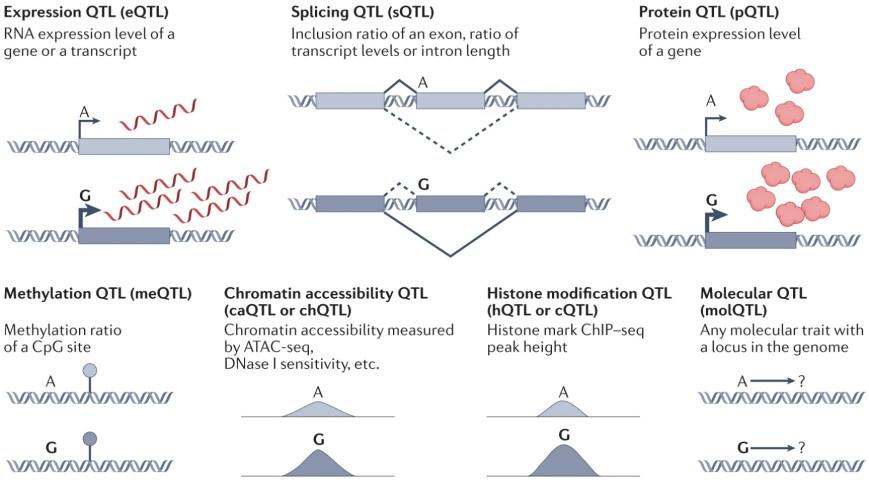



**Figure:** Aguet *et al*. (2023) Nature Reviews Methods Primers

# Regulatory variation in complex traits and disease

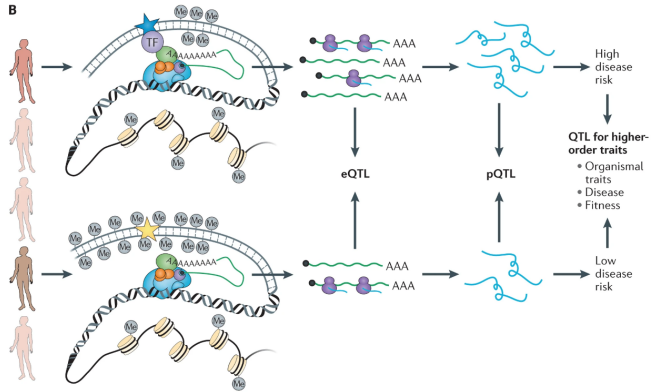



**Figure:** Albert & Kruglyak (2015) Nature Reviews Genetics

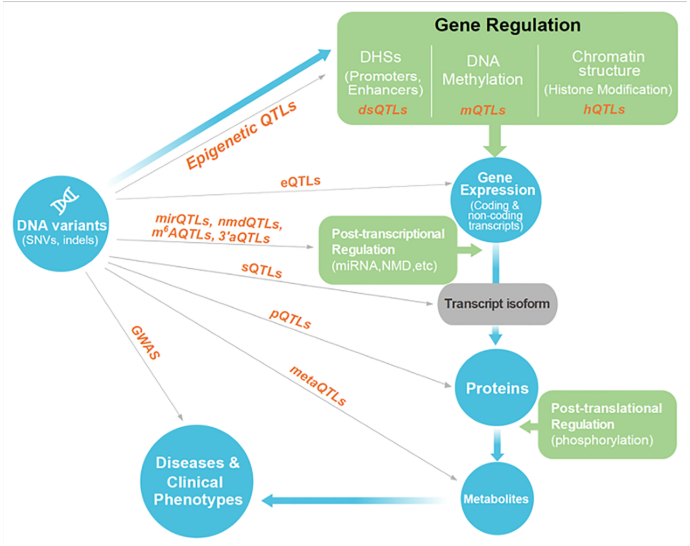



**Figure:** Olayinka *et al*. (2022) Curr Protoc.

# Molecular QTL (xQTL) Categories

## DNA and Chromatin Modifications

**DNA Methylation**
meQTL (methylation QTL)
dmrQTL (differentially methylated region QTL)

**Chromatin State**
haQTL (histone acetylation QTL)
tfbsQTL (transcription factor binding site QTL)
caQTL (chromatin accessibility QTL)

## RNA Expression and Processing

**Gene Expression**
eQTL (expression QTL)
ct-eQTL (cell-type specific eQTL)
aseQTL/aimQTL (allele-specific expression QTL)
lncQTL (long non-coding RNA QTL)
mircQTL (microRNA QTL)
circQTL (circular RNA QTL)
piQTL (pi RNA QTL)

**RNA Processing**
sQTL (splicing QTL: intron usage, exon skipping, splice site usage)
atsQTL (alternative transcript start QTL)
apaQTL (alternative polyadenylation QTL)
nmdQTL (nonsense-mediated decay QTL)
iso-eQTL (isoform-specific QTL)
m6A-QTL (RNA modification QTL)

## Protein and Other Molecular Phenotypes

**Proteomics** | pQTL (protein QTL)
**Metabolomics & Lipidomics** | metaQTL (metabolite QTL)

# Functional genomics context of xQTL

- Functional genomics features
  - cis-regulatory elements, histone marks, TF binding sites, chromatin accessibility, chromatin structure (Hi-C), etc.
  - Tissue / cell specific functional genomics contexts
- Other coding, conserved, MAF and LD-related annotations
- Machine learning derived

# Other xQTL contexts

**Multiple tissues**

- Various brain tissues / cells
- CSF
- Other relevant tissues / cells (e.g. immune response)

**Multi-ancestry & cohorts**

- European
- African / African American
- Hispanic

**Multiple, related complex traits**

- Alzheimer's disease (AD), AD endophenotypes, related neurodegenerative disorders

# cis- (local) and trans- (distant) xQTL

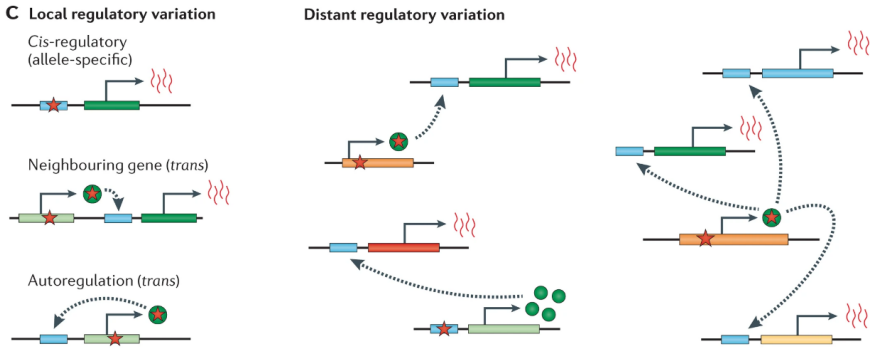



**Figure:** Albert & Kruglyak (2015) Nature Reviews Genetics

# eQTL mapping workflow and example

Credit: eQTL mapping workflow inspired by Dr. Xin He's lecture notes
sn-eQTL example illustrates Dr. Masashi Fujita's work, Fujita *et al*. 2024.

# RNA-seq measures gene expression

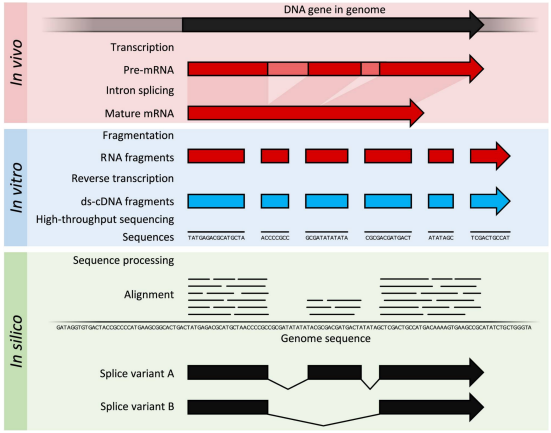



**Figure:** Lowe *et al*. (2017) PLoS Comp. Bio.

# eQTL mapping: challenges compared to GWAS

Expression quantative trait loci (eQTL) mapping

- Identifies genetic variants associated with gene expression, revealing regulatory mechanisms.
- Challenges:
  - Noisy gene expression data influenced by sample quality and cell type composition.
  - Need for rigorous control of false discoveries in large datasets.
  - High dimensionality due to testing many genes against numerous variants.

# Procedure of xQTL mapping: an overview

- **Quantification of molecular traits.** Such as expression levels, splicing patterns, methylation, and protein abundance, measured using technologies like RNA-seq, bisulfite sequencing, or proteomics.
- **QC and normalization of molecular traits.** Such as quantile normalization adjust for systematic biases, enabling comparisons across samples.
- **Identify hidden covariates.** Hidden factors such as cell-type composition and batch effects are addressed using techniques like PCA or factor analysis.
- **Genotype data QC and preprocessing.** Similar to GWAS.
- **Missing molecular phenotype data imputation.**
- **Association mapping for thousands of molecular features.** Identify genetic associations with molecular traits: simple regression with multiple testing correction or variable selection regression such as fine-mapping.

# Quantifying gene expression levels from read counts

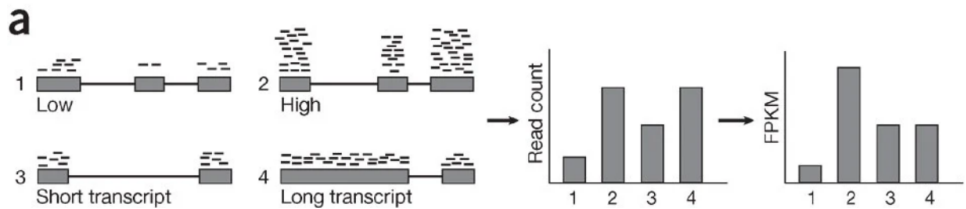



**Figure:** Carber *et al*. (2011) Nature Methods

- The number of reads mapping to a transcript reflects its expression level.
- However, read counts also depend on:
  - Gene length
  - Total number of reads in a sample (library size)

# Common measures of gene expression from RNA-seq

- **RPKM** (Reads Per Kilo-base exon model per Million mappable reads) $R = \frac{10^9 \cdot C}{N \cdot L}$, where:
  - $C$: number of reads in a transcript
  - $N$: total number of reads in a sample
  - $L$: gene length
- Not suited for eQTL mapping: sum of RPKM reads in each sample can be different.
- **TPM** (Transcripts Per Million) Similar to RPKM, but total TPMs across genes sum to a constant ($10^6$).
- $\text{Log}_2$ transformation is often applied to RPKM/TPM values.

# Gene expression data often needs additional normalization

- Goal: Adjust systematic biases for easier comparison across samples.
- Library size difference is one bias addressed by RPKM/TPM normalization.
- Additional sources of bias:
  - Sequencing bias toward GC-rich genes.
  - Dependency among genes due to RNA-seq protocol: higher expression of some genes reduces read counts of others.
- Other normalization methods mitigate these biases.
  - `DESeq2` uses median ratios of $\frac{\text{gene count in sample}}{\text{geometric means across samples of the gene}}$ across samples to adjust both library size and composition biases.

# Quantile normalization is a general strategy to adjust for sample differences

- Systematic differences in samples arise from multiple sources.
- Library size and composition biases adjustments may not suffice.
- Quantile normalization adjusts data so that *gene expression distributions are identical across samples*.
- Genes at **the same rank** in different samples are **assigned the same expression value** post-normalization.

# Procedure of quantile normalization

- For a gene in sample $j$, let the quantile of its expression be $q$ and its expression level be $x_{qj}$.
- Adjust expression to the mean expression of genes at rank $q$ across samples:

$$\tilde{x}_q = \frac{1}{n} \sum_{j=1}^{n} x_{qj}$$

- Replace $x_{qj}$ in sample $j$ with $\tilde{x}_q$.

# Association is typically tested by linear regression

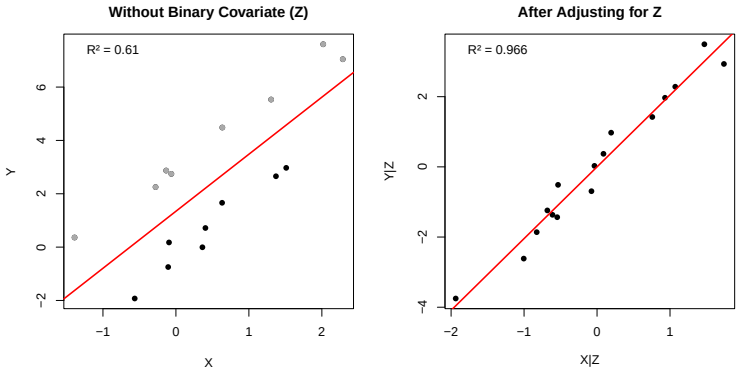
- Linear model tests the association between normalized expression $Y$ and SNP genotypes $X$, adjusting for covariates $Z$:

$$Y = X\beta + Z\gamma + \varepsilon$$

- Covariates:
  - Known: age, gender, RNA quality, ancestry, batch effects.
  - Hidden: cell growth rate, unmeasured batch effects, cell type composition.

# Adjusting for covariates in association testing increases power

Regressing out $Z$ improves power for detecting the association between $X$ and $Y$.



# Adjusting for covariates in association testing example

```
n <- 16; X <- rnorm(n)
Z <- rep(c(0,1), each=n/2)  # Binary covariate
Y <- 2*X + 3*Z + rnorm(n, sd=0.3)  # Strong Z effect
mod1 <- lm(Y ~ X)  # Without covariate
Xresid <- residuals(lm(X ~ Z))  # Adjust X for Z
Yresid <- residuals(lm(Y ~ Z))  # Adjust Y for Z
mod2 <- lm(Yresid ~ Xresid)  # After adjustment
r2 <- c(raw = summary(mod1)$r.squared,
  adjusted = summary(mod2)$r.squared)
```

# PCA is often used to adjust hidden covariates

Hidden covariates, such as cell type composition, often influence the expression of **many genes**. We can leverage gene expression correlations to identify and adjust for these hidden factors.

- **PCA (Principal Component Analysis)**:
  - Projects expression data into lower dimensions.
  - Each sample is represented by a small number of principal component (PC) scores.
  - These PC scores are used as covariates in QTL association tests.

# PCA is often used to adjust hidden covariates

Hidden covariates, such as cell type composition, often influence the expression of **many genes**. We can leverage gene expression correlations to identify and adjust for these hidden factors.

- Number of PC to include can be estimated from data:

**PCA outperforms popular hidden variable inference methods for molecular QTL mapping**

Heather J. Zhou, Lei Li, Yumei Li, Wei Li & Jingyi Jessica Li

*Genome Biology* **23**, Article number: 210 (2022) | Cite this article

# Multiple testing correction in cis-eQTL mapping

Statistical significance of xQTL:

- For cis-xQTL mapping, control the False Discovery Rate (FDR) at the gene level.
- SNPs in linkage disequilibrium (LD) often cluster near a gene, so reporting all significant SNPs is not efficient.

Identifying **eGenes:**

- eGenes are defined as genes with at least one significant cis-eQTL.
- Null hypothesis: No SNP near a gene is associated with its molecular trait.
- A gene-level statistic (e.g., minimum p-value of SNPs) is used for further analysis.

# Multiple testing correction in cis-eQTL mapping

Step 1: permutation testing for null distribution **at one gene:**

- Generate the null distribution by permuting sample labels and recomputing gene-level statistics (e.g. minimum p-value).
- Calculate empirical p-values by comparing observed statistics to the null.

$$p_{\text{empirical}} = \frac{\#(\text{null test statistics} \geq \text{observed statistic})}{\text{total permutations}}$$

- Tools like `TensorQTL` perform permutation tests efficiently ("adaptive permutation").

# Multiple testing correction in cis-eQTL mapping

Step 2: FDR for **genome-wide significance:**

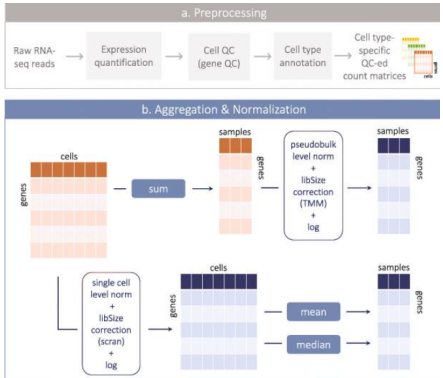
- Apply FDR correction to empirical p-values to identify significant eGenes genome-wide.
- BH procedure, or q-value

# Multiple testing correction in cis-eQTL mapping

**Bayesian fine-mapping** provides an alternative to multiple testing control:

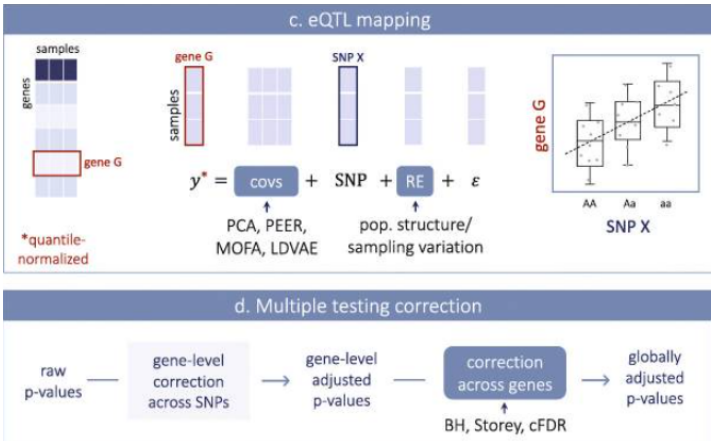
- Posterior inclusion probability (PIP) and credible sets (CS) are used to assess SNP significance.
- Multiple testing correction is unnecessary because only one test is performed per gene, modeling all SNPs together with PIP directly quantify SNP relevance.
- eGenes can be defined as those with at least one credible set of fine-mapped SNPs.

# Cell type eQTL workflow: expression quantification




Cuomo *et al*. (2021) Genome Biol.

# Cell type eQTL workflow: association testing




Cuomo *et al*. (2021) Genome Biol.

# A case study: cell specific eQTL in brains

single-nuclei RNAseq (snRNAseq) from frozen human brain in the prefrontal cortex region

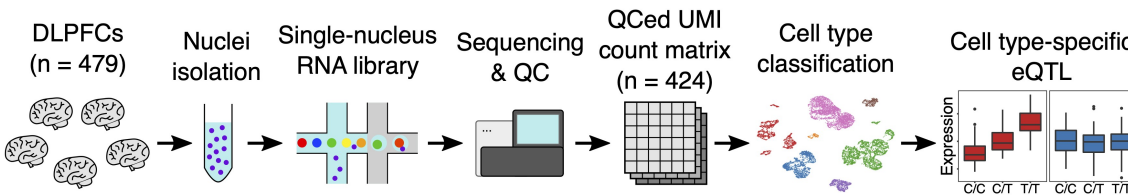
Article | Published: 21 March 2024

## Cell subtype-specific effects of genetic variation in the Alzheimer's disease brain

Masashi Fujita, Zongmei Gao, Lu Zeng, Cristin McCabe, Charles C. White, Bernard Ng, Gilad Sahar Green, Orit Rozenblatt-Rosen, Devan Phillips, Liat Amir-Zilberstein, Hyo Lee, Richard V. Pearse II, Atlas Khan, Badri N. Vardarajan, Krzysztof Kiryluk, Chun Jimmie Ye, Hans-Ulrich Klein, Gao Wang, Aviv Regev, Naomi Habib, Julie A. Schneider, Yanling Wang, Tracy Young-Pearse, Sara Mostafavi, ... Philip L. De Jager ✉ + Show authors

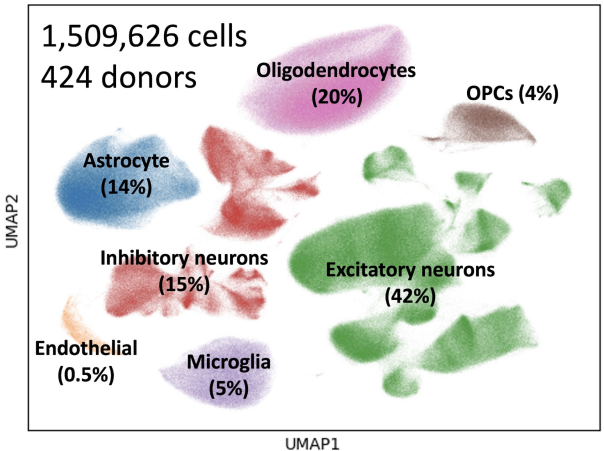
*Nature Genetics* **56**, 605–614 (2024) | Cite this article

# Study design overview

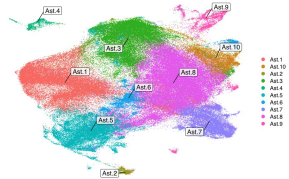



- Frozen dorsolateral prefrontal cortex (DLPFC) tissues of ROSMAP cohort were analyzed.
- ROSMAP participants have agreed to receive annual cognitive testing and to donote their brain upon their death. Donated brains were pathologically examined.
- Single cell gene expression libraries were constructed using 10x Genomics 3’ kit.

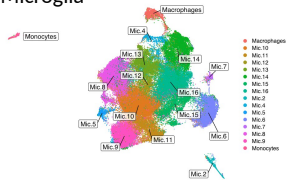
# Atlas of brain cells in DLPFC



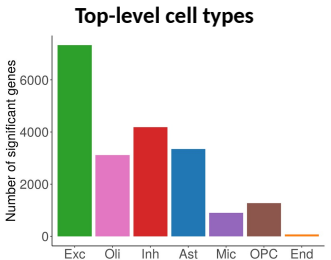

Astrocytes

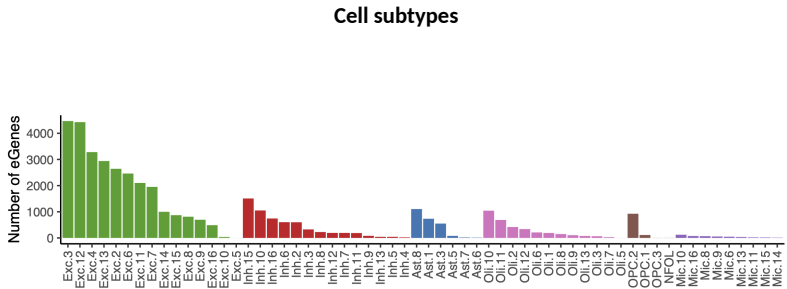



Microglia



# Cell specific gene and eQTL (ct-eQTL)

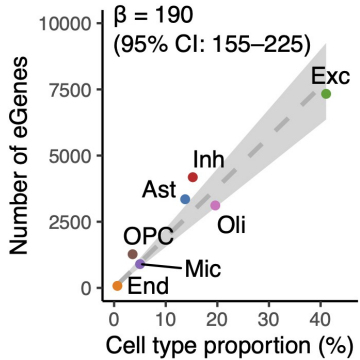


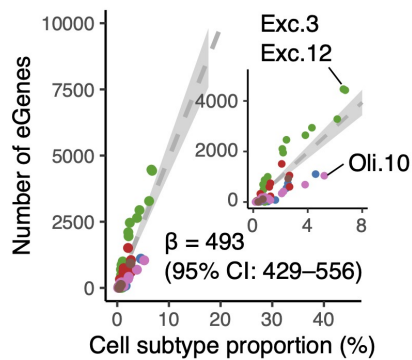
# Limitation: rare cell types are under-represented

Cell type proportion is correlated with eQTL detection power

**Cell types**

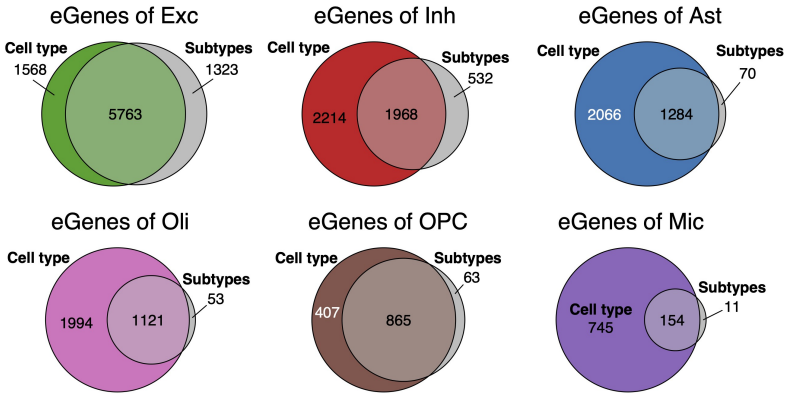



**Subtypes**

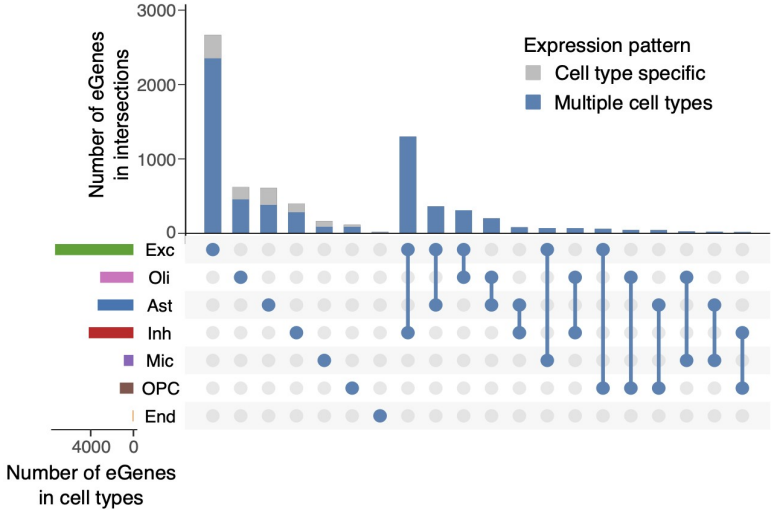


# ct-eQTL discovery and comparison with eQTL

Subtype eQTL analysis boosted eQTL detection

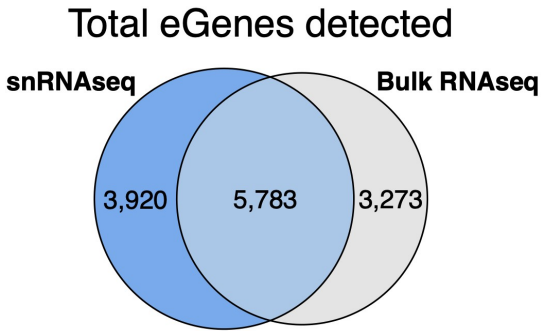


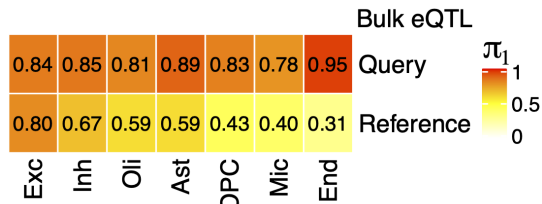
# Summary of ct-eQTL specificity



# Replication in bulk RNAseq eQTL

Low replication rate $\pi_1$ in rare cell types is consistent with the lack of detection power in those cell types

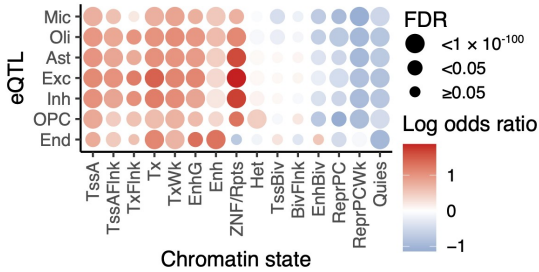


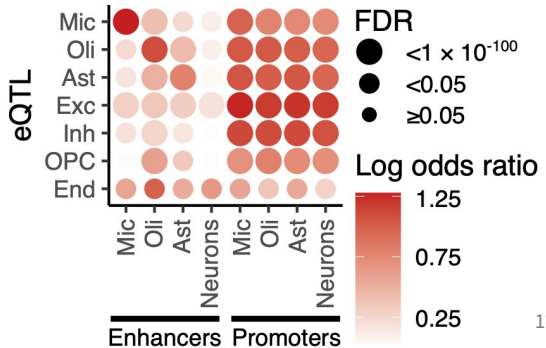
# Chromatin states of ct-eQTL

eQTL are enriched in euchromatin (transcriptionally active regions and enhancers) and relatively depleted in heterochromatin

**Chromatin states in bulk DLPFC**

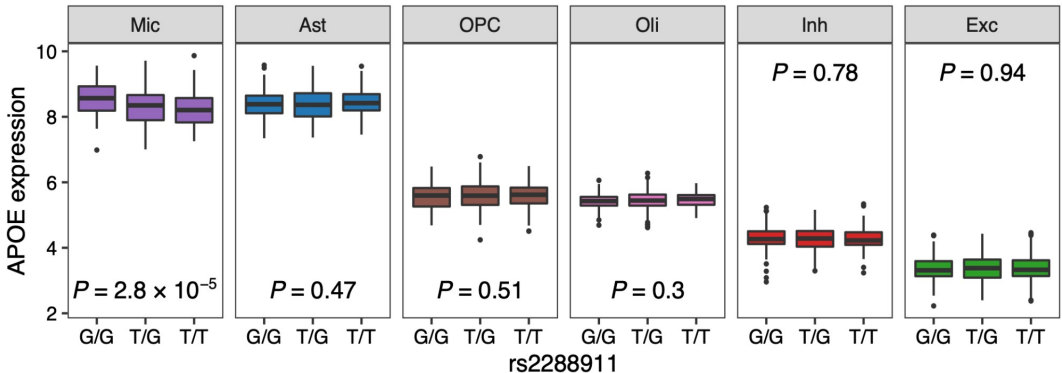



**Brain cell type-specific enhancers and promoters (Nott et al., 2019)**



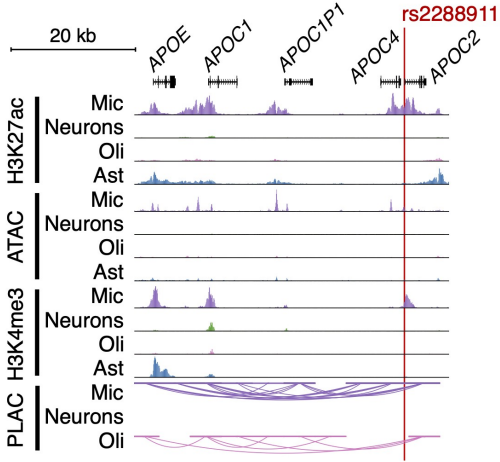
# A discovery relevant to Alzheimer's disease

*APOE* (Apolipoprotein E) expression is specific to Mic and Ast; it's genetic regulation is specific to Mic.



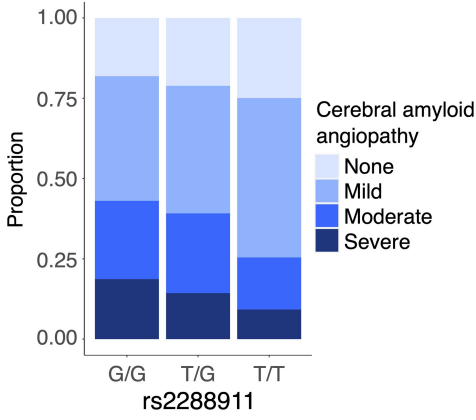
# Follow up on Mic. specific eQTL rs2288911

rs2288911 is located at a microglia-specific enhancer

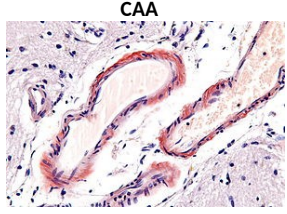


# Follow up on Mic. specific eQTL rs2288911

rs2288911 is a risk SNP of AD and cerebral amyloid angiopathy (CAA)

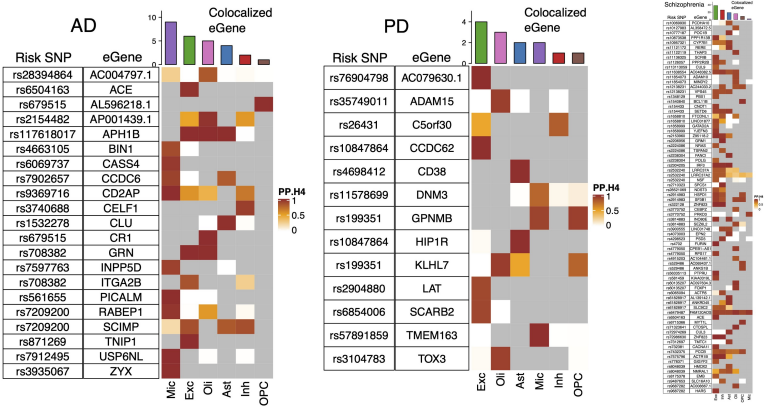



- rs2288911 is associated with the risk of AD in GWAS ($p = 1.0 \times 10^{-12}$).
- In ROSMAP, this SNP is associated with CAA ($p = 1.2 \times 10^{-7}$). This association persists even after adjusted for the *APOE* ε4 allele.



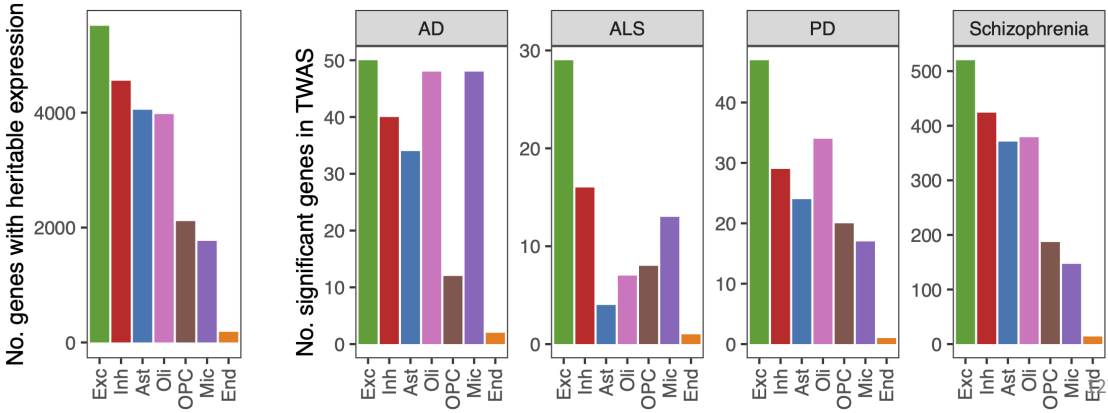
# Integrative study with disease GWAS

Colocalization with GWAS signals of neurological diseases highlights the necessity of ct-eQTL studies



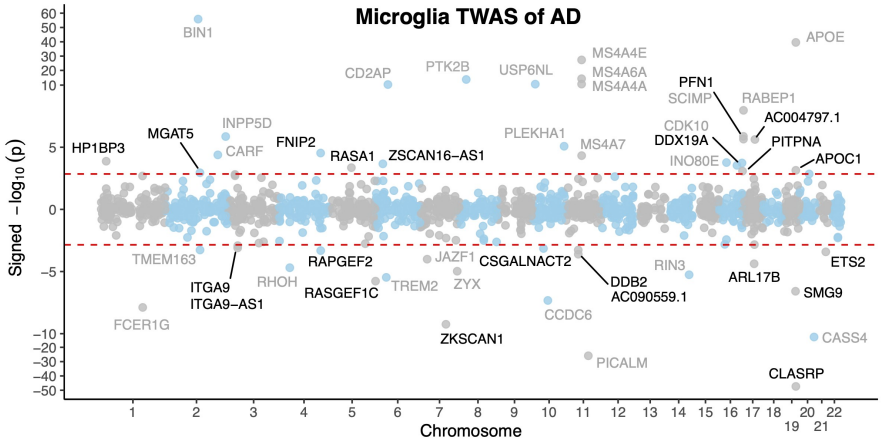
# Transcriptome-wide association studies

Cell type-specific TWAS shows enrichment of microglia genes in Alzhemier's disease



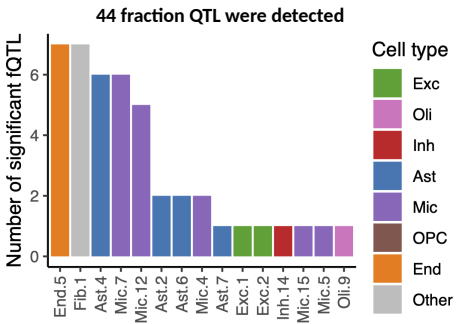
# Transcriptome-wide association studies

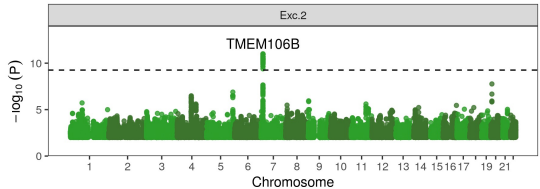
Microglia-specific TWAS identifies novel genes associated with Alzhemier's disease

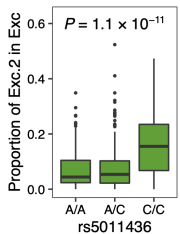


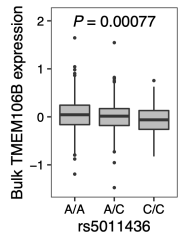
# QTL of cell fraction may confer disease risk

Fraction QTL (fQTL) of Exc.2 colocalizes with AD risk SNP of gene *TMEM106B*



# Rare xQTL can improve PRS for complex traits

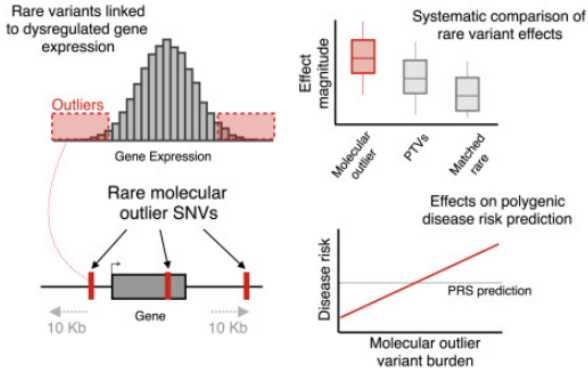



**Figure:** Smail *et al*. (2022) AJHG

Also see Li *et al*. (2017) Nature; Ferraro *et al*. (2020) Science

# Missing regulation in eQTL and GWAS

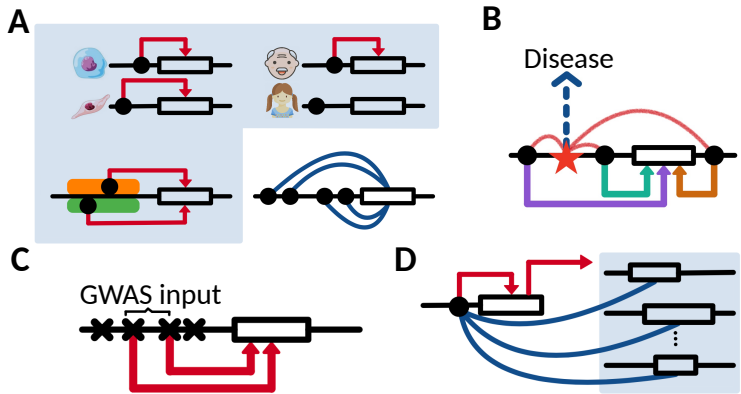
## The missing link between genetic association and regulatory function

Noah J Connally, Sumaiya Nazeen, Daniel Lee, Huwenbo Shi, John Stamatoyannopoulos, Sung Chun, Chris Cotsapas, Christopher A Cassa, Shamil R Sunyaev

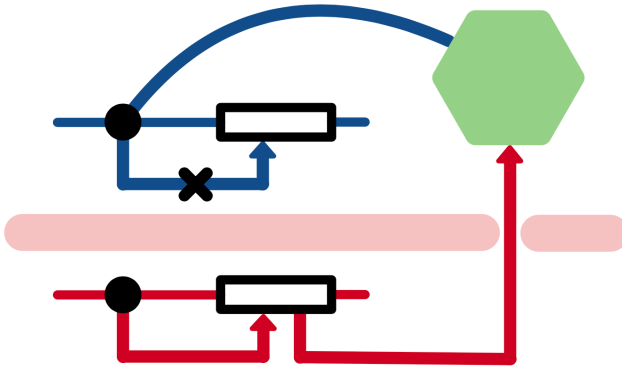
… by applying a gene-based approach we found limited evidence that the baseline expression of trait-related genes explains GWAS associations, whether using colocalization methods (8% of genes implicated), transcription-wide association (2% of genes implicated), or a combination of regulatory annotations and distance (4% of genes implicated). These results contradict the hypothesis that most complex trait-associated variants coincide with homeostatic expression QTLs, suggesting that better models are needed. The field must confront this deficit and pursue this 'missing regulation.'

Connally et al, December 2022, elife; also see Mostafavi et al + Prichard 2022

# Fancier methods, or more targeted hypotheses?



# Fancier methods, or more targeted hypotheses?



# QTL-GWAS loci: multi-trait analysis and colocalization

# More phenotypes, more complications

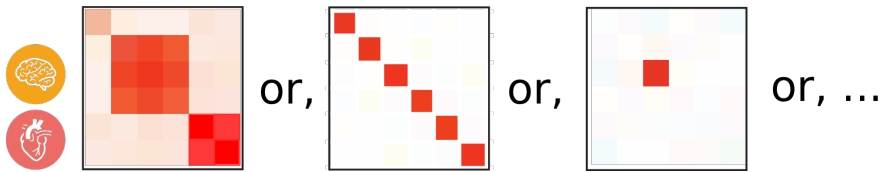



**Figure:** Plausible patterns of sharing

# Major challenges

- **For a given variant**: the less assumption made on multivariate effects, the more parameters to estimate.
  - Fixed effect (FE) and Random Effects (RE) models are restrictive but easy to fit.
- **Different variants**: may fit in different multivariate effect models, **local patterns of genetic heterogeneity among traits**

# A naive mixture model

"FE and RE are equally likely for any variant":

$$U_{mixed} = 0.5 \times \begin{bmatrix} \sigma_0^2 & \sigma_0^2 \\ \sigma_0^2 & \sigma_0^2 \end{bmatrix} + 0.5 \times \begin{bmatrix} \sigma_0^2 & 0 \\ 0 & \sigma_0^2 \end{bmatrix}$$

Prior allows for equal possibility of both; data will determine where posterior lands.

# A data-adaptive mixture model

Instead of making assumptions, can we **learn from data**:

- What are the latent structures for multivariate effects?
- How often does each structure appear?

and use these to construct the mixture model?

# What are shared, and what ain't?

Decomposing effect estimates (gene by tissue), $\widehat{\beta} = \boldsymbol{LF} + \boldsymbol{E}$

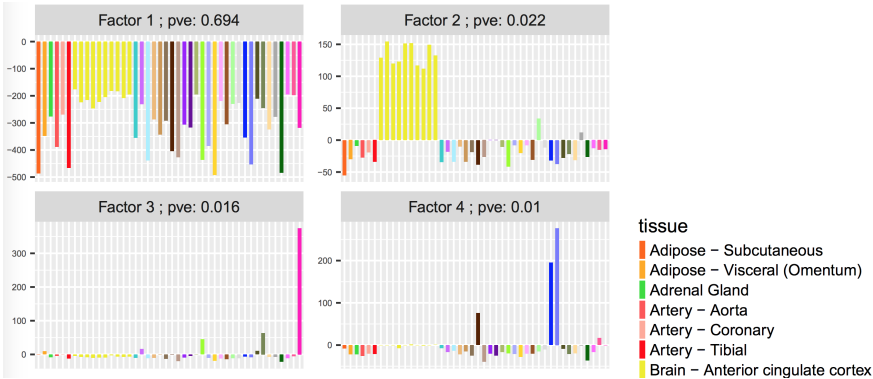



**Figure:** Sparse factor analysis of GTEx data

# Incorporating all possible patterns

Multivariate effects of a variant follows the $k$-th pattern with probability $\pi_k$:

$$U_{mixed} = \pi_1 \times \begin{bmatrix} 2.4 & 0.3 \\ 0.3 & 1.5 \end{bmatrix} + \pi_2 \times \begin{bmatrix} 1.6 & 0.001 \\ 0.001 & 0.02 \end{bmatrix} + \pi_3 \times \cdots$$

# How much are shared?

**M**ultivariate **A**daptive **SH**rinkage model for effect estimates:

$$\widehat{\boldsymbol{\beta}} \sim N_R(\boldsymbol{\beta}, \boldsymbol{S})$$
$$\boldsymbol{\beta} \sim \sum_k \pi_k N_R(\mathbf{0}, \boldsymbol{U}_k)$$

$\boldsymbol{U}_k$ was **learned from data**, $\pi_k$ is **to be learned** next!

# How much are shared?

Maximum likelihood for the finite mixture model:

$$\widehat{\pi} = \text{argmax} \sum_j \log\left(\sum_k \pi_k L_{k_j}\right)$$

This is a convex optimization problem $\Rightarrow$ **very fast to fit!**

# How much are shared?

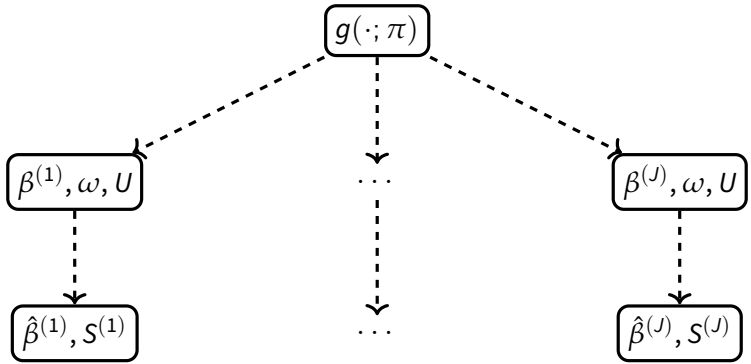
Maximum likelihood for the finite mixture model:

$$\widehat{\pi} = \text{argmax} \sum_j \log\left(\sum_k \pi_k L_{k_j}\right)$$

Multiple testing is **an opportunity to learn!**

- The more the tests, the better we learn.

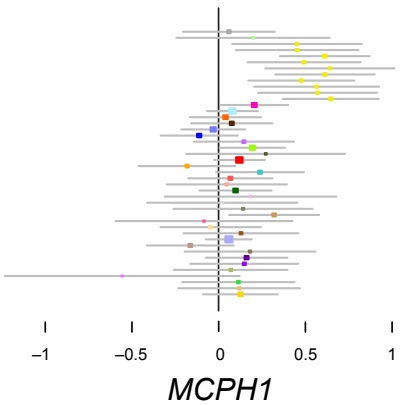
# Multivariate Adaptive SHrinkage Illustrated



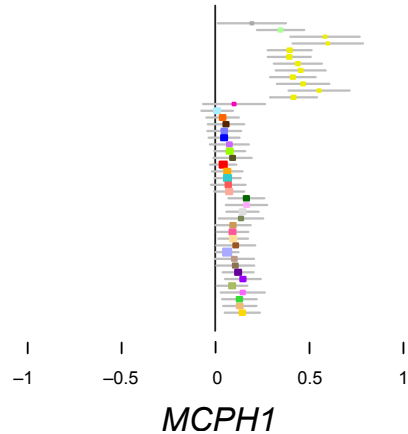

- Step 1: estimated $\pi_k$ via EM algorithm using data across genome.
- Step 2: apply this prior to each variant in association mapping.

# *MASH* improves power

Original effect estimates

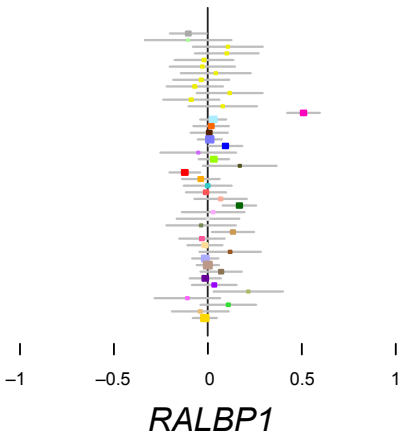



*MASH* effect estimates

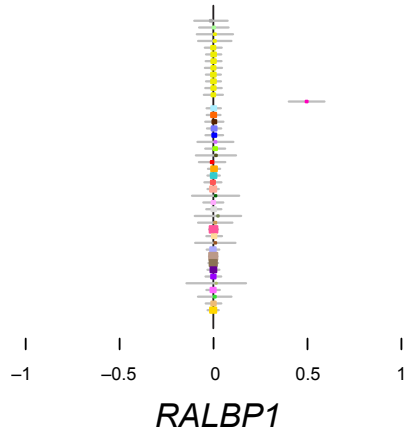


# *MASH* reduces noise

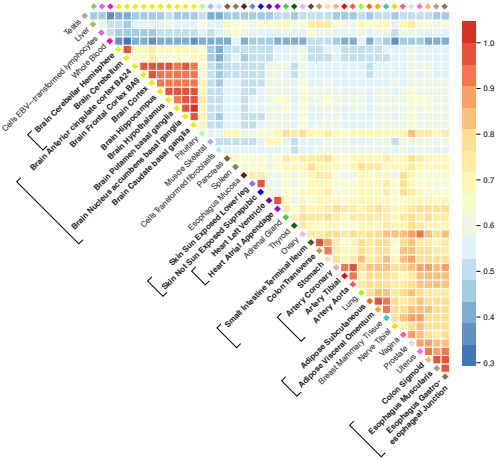
Original effect estimates




*MASH* effect estimates



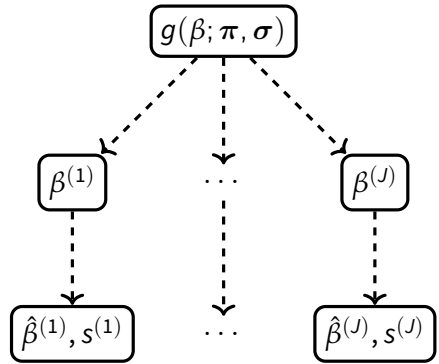
# *MASH* reveals quantitative heterogeneity



# A genome-wide *SuSiE* model

Learn shrinkage priors from genome-wide univariate tests

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{b} + \boldsymbol{e}$$
$$\boldsymbol{e} \sim N(0, \sigma^2 I_n)$$
$$\boldsymbol{b} = \sum_{l=1}^{L} \boldsymbol{b}_l$$
$$\boldsymbol{b}_l = \gamma_l \beta_l$$
$$\gamma_l \sim \text{Mult}(1, \boldsymbol{\alpha})$$
$$\beta_l \sim \sum_k \pi_k N(0, \sigma_k^2)$$



# Multivariate Multiple Regression fine-mapping

The *mvSuSiE* model,

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{B} + \boldsymbol{E}$$

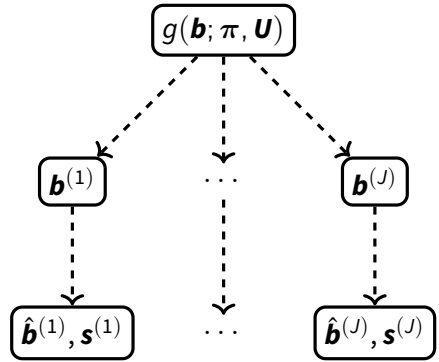
$$\boldsymbol{E} \sim N_{N\times R}(\boldsymbol{0}, \boldsymbol{I} \otimes \boldsymbol{V})$$

$$\boldsymbol{B} = \sum_{l=1}^{L} \boldsymbol{B}_l$$

$$\boldsymbol{B}_l = \gamma_l \boldsymbol{b}_l^{\mathsf{T}}$$

$$\gamma_l \sim \text{Mult}(1, \boldsymbol{\alpha})$$

$$\boldsymbol{b}_l \sim \sum_{k,m} \pi_{km} N_R(\boldsymbol{0}, \omega_m \boldsymbol{U}_k)$$



# Application to multivariate fine-mapping

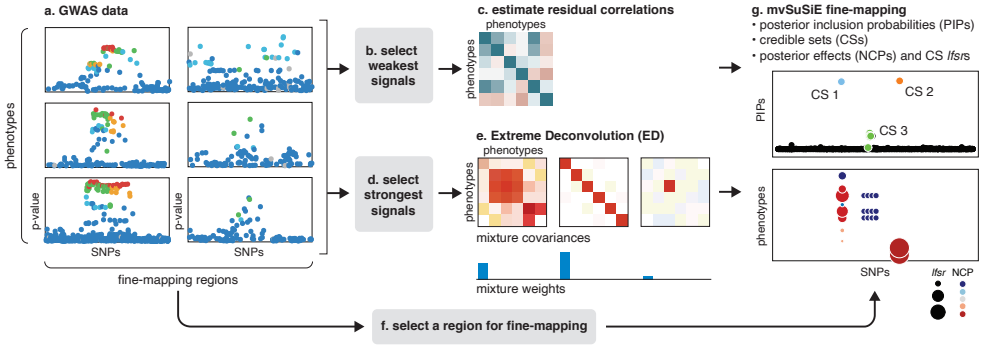



**Figure:** mvSuSiE fine-mapping with adaptive shrinkage model

Zou *et al*. (2023) biorxiv

# Multi-ancestry fine-mapping as a special case

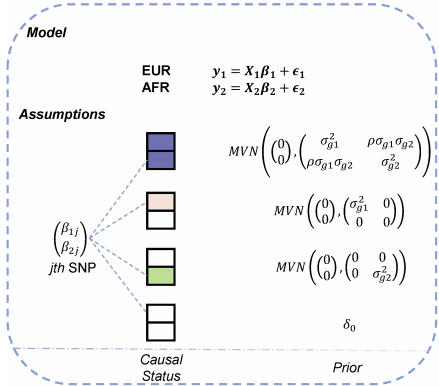
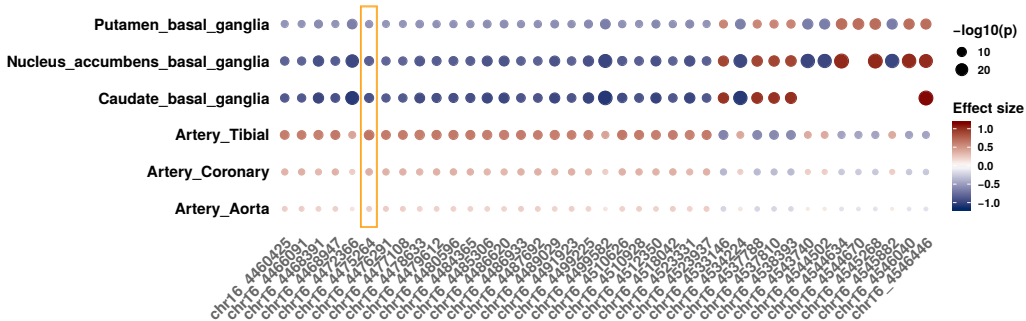


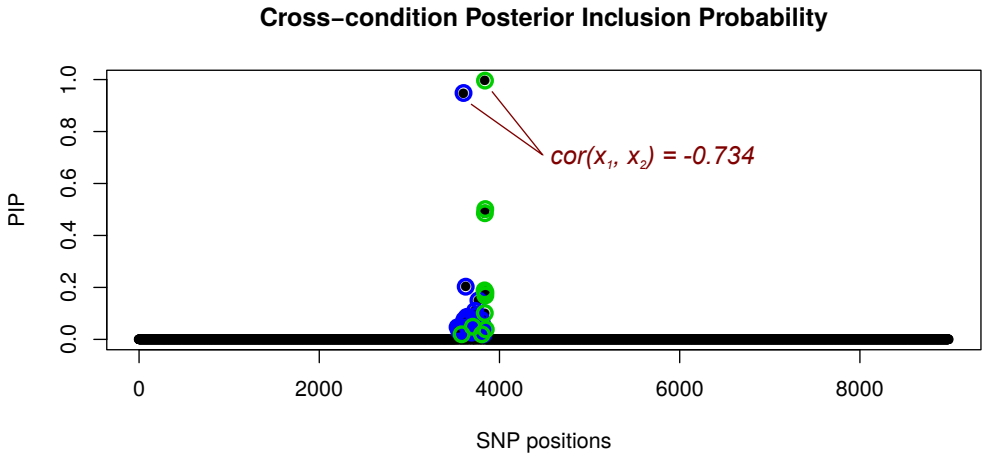

**Figure:** MESuSiE for cross-ancestry fine-mapping

Gao & Zhou *et al*. (2024) Nat. Genet.

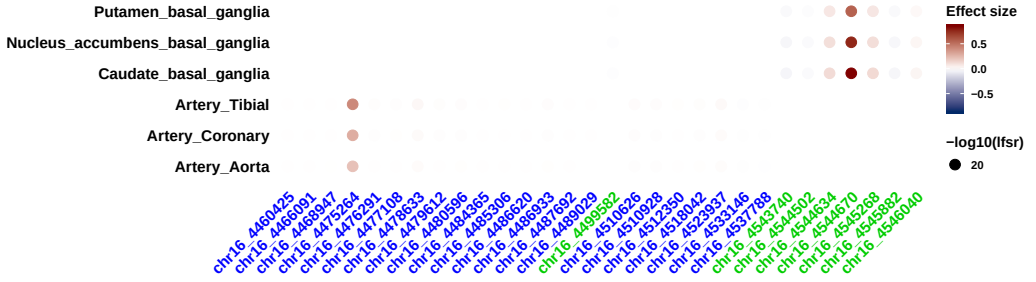
# “one eQTL per gene” architecture



# *mvSuSiE* fine-mapping architecture



# *mvSuSiE* fine-mapping architecture

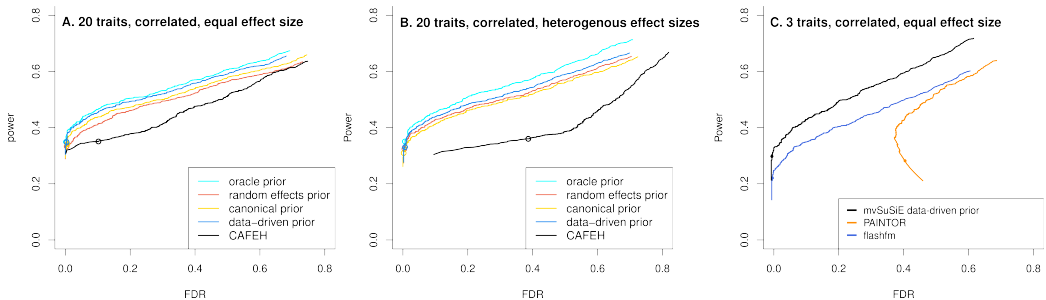


# Multi-trait fine-mapping methods & challenges

| | mvSuSiE | CAFEH | PAINTOR | MTHESS | BayesSUR | flashfm | msCaviar | HyPrColoc | moloc |
|---|---|---|---|---|---|---|---|---|---|
| >5 traits integrated | ■ | ■ | | | | | | ■ | |
| >10 traits integrated | ■ | ■ | | | | | | ■ | |
| Multiple causal signals | ■ | ■ | | ■ | ■ | | | | |
| Individual level data | ■ | ■ | | ■ | ■ | | | | |
| Summary statistics | ■ | ■ | ■ | | | ■ | ■ | ■ | ■ |
| Missing data | | | | | | ■ | | | |
| Trait specific LD | | | | | | | ■ | | |
| Correlated effects | ■ | | | ■ | ■ | ■ | ■ | | |
| Trait specific effects | ■ | ■ | | | | ■ | | ■ | ■ |
| Arbitrary heterogeneous effects | ■ | | | | | ■ | | | |
| Arbitrary multi-trait colocalization | | ■ | | | | ■ | | ■ | ■ |
| Correlated traits | ■ | | | ■ | ■ | ■ | | ■ | |
| Partial sample overlap | ■ | | | | | ■ | | ■ | |
| Functional annotation | | | ■ | | | | | | |
| Trait specific functional annotation | | | | | | | | | |
| Genome-wide scalability | ■ | ■ | | | | | | ■ | |

Reference: CAFEH: Arvanitis et al (2022); PAINTOR: Kichaev et al (2017); MTHESS: Lewin et al (2016); BayesSUR: Zhao et al (2021); flashfm: Hernández et al (2021); msCaviar: LaPierre et al (2021); HyPrColoc: Foley et al (2021); moloc: Giambartolomei et al (2018).

# Comparison to other methods



# GWAS application: 16 blood traits in UK Biobank

Analysis overview

- Sample size 248,980; 975 candidate regions fine-mapped
- Average #SNPs per region 4,776; maximum 36,605

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Analysis overview

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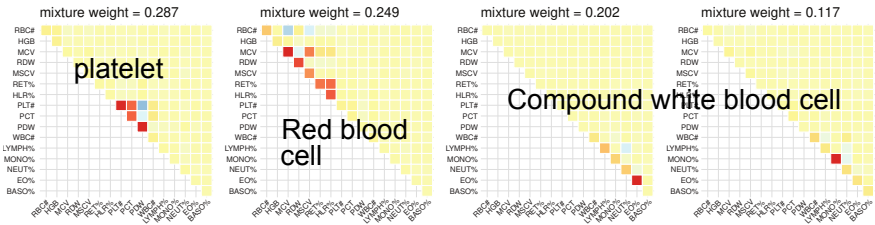
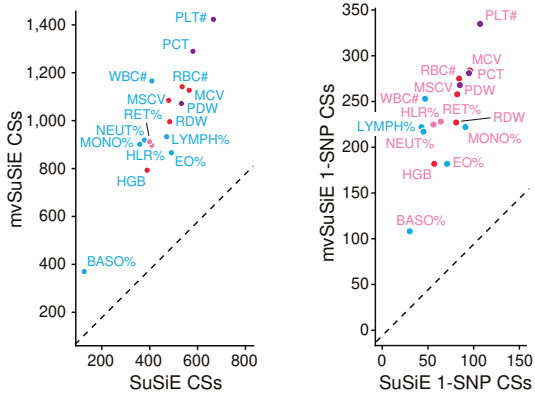
Top patterns of effect size sharing inferred from data:



# GWAS application: 16 blood traits in UK Biobank

Many more signals identified compared to fine-mapping per each trait



# Beyond per trait per variant association studies

**Statistical fine-mapping (multiple regressors)**

- Identify non-zero effect variables by accounting for LD

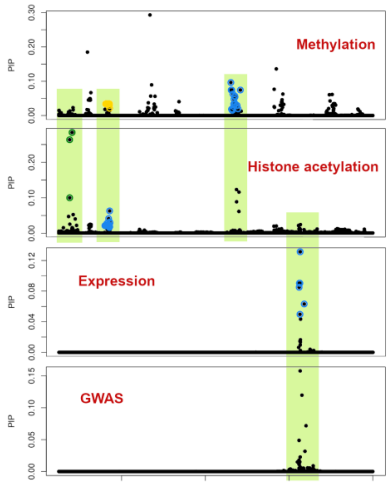
**Meta-analysis (multiple responses)**

- Integrate information across multiple conditions / studies

**“Causal” variants across multiple conditions?**

- Cross-population fine-mapping; colocalization; pleiotropy; mediation; …

# The problem


Methylation
Histone acetylation
Expression
GWAS
PIP
SNPs within 500Kb radius of *ABCA7* transcription start site

# The problem

For a genetic variable analyzed in two conditions:

$$P(\text{“causal” in trait 1 \& 2} \mid \text{association data for 1 \& 2})$$

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$$P(\text{“causal” in trait 1 \& 2} \mid \text{association data for 1 \& 2})$$

Denote data as $D_1$ and $D_2$, and use indicator variables $\gamma_1, \gamma_2$ for variable having effects in 1 and 2, and hyperparameters $\Theta$:

$$P(\gamma_1 = 1, \gamma_2 = 1 | D_1, D_2, \Theta)$$

# Multivariate relationships?

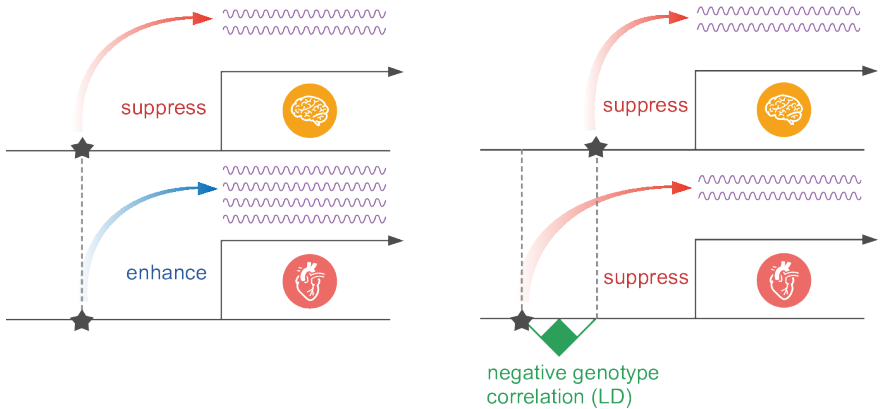


**Figure:** Pleiotropy or Linkage?

# QTL-GWAS colocalization model and inference

- **Problem:** For variants analyzed in GWAS and eQTL studies, compute:

$$P(\gamma_g = 1, \gamma_e = 1 | D_g, D_e, \Theta)$$

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  - $H_0$: No associations ($\gamma_g = 0, \gamma_e = 0$)
  - $H_1$: eQTL only ($\gamma_g = 0, \gamma_e = 1$)
  - $H_2$: GWAS only ($\gamma_g = 1, \gamma_e = 0$)
  - $H_3$: Two independent variants ($\gamma_g = 1, \gamma_e = 1$, independent)
  - $H_4$: Shared causal variant ($\gamma_g = 1, \gamma_e = 1$, colocalized)

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  - $H_4$: Shared causal variant ($\gamma_g = 1, \gamma_e = 1$, colocalized)
- **Posterior computation:**

$$PP_h \propto \sum_{S \in S_h} P(S) \cdot P(D|S)$$

where $S_h$ are configurations for model $H_h$, with **prior** $P(S)$ and likelihood $P(D|S)$

# Colocalization method: *coloc*

*coloc* [Giambartolomei *et al*. (2014) PLoS Genet.]

- On *X*: “one causal” assumption
- On *Y*: the null + 4 combinations given “one causal“
  1. In 1 but not 2
  2. In 2 but not 1
  3. In 1 and 2 but not the same variable
  4. In 1 and 2 and the same variable (colocalization)
  5. No association in both data 1 and 2

# Colocalization method: *eCAVIAR*

*eCAVIAR* [Hormozdiari *et al*. (2016) Am. J. Hum. Genet.]

- On $X$: multiple effect variables
- On $Y$: each effect variable can be
  1. In 1 but not 2
  2. In 2 but not 1
  3. In both 1 and 2
  4. No association in both data 1 and 2

# *eCAVIAR* effects assumption

Effect sizes are independent,

$$U = \begin{bmatrix} \sigma_g^2 & 0 \\ 0 & \sigma_e^2 \end{bmatrix}$$

# Colocalization method: *enloc*

*enloc* [Wen *et al*. (2017) PLoS Genet.]

- Key difference: cross-condition effects **not** independent
- **eQTL signals are enriched in GWAS**

# Colocalization method: *enloc*

*enloc* [Wen *et al.* (2017) PLoS Genet.]

- Key difference: cross-condition effects **not** independent
- **eQTL signals are enriched in GWAS**

But how?

- Through a simple logistic link **using eQTL as an annotation** for $j$

$$\log\left[\frac{\pi}{1-\pi}\right] = \alpha_0 + \alpha\gamma_e$$

and in this context

$$\pi := P(\gamma_g = 1 | \gamma_e = 1)$$

# *enloc* two step procedure

1. Obtain $P(\gamma_g = 1)$ and $P(\gamma_e = 1)$ using fine-mapping
2. Fit the enrichment model via **multiple imputation**

# Connections between colocalization methods

- *eCAVIAR* is a special case of *enloc* with $\alpha = 0$.
- *coloc* is a special case of "one causal" fine-mapping based *enloc* with fixed, high(!) $\alpha$ value by default.
- Recent *coloc* extension: *coloc* version 5, aka *SuSiE-coloc* removed the "one causal" assumption.
  - Wallace (2021) PLoS Genetics
  - `https://chr1swallace.github.io/coloc/`

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Summary: **pattern** and **scale** of effect size correlations, represented as different **prior** models.

# Practical considerations

- Choice of prior
  - Best to **estimate enrichment $\alpha$ from data**
  - $\alpha \in [0, 5]$ suggested by $> 4,000$ GWAS + GTEx data
- LD reference mismatch: underestimate $\alpha$, thus power loss

Hukku *et al.* (2021) Am. J. Hum. Genet.

# Multi-trait colocalization model configurations

Assuming a single causal variant in the loci.

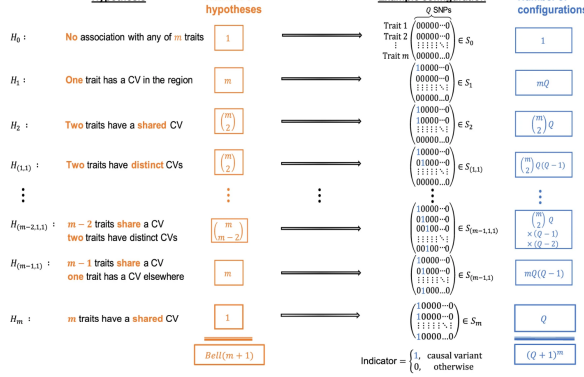
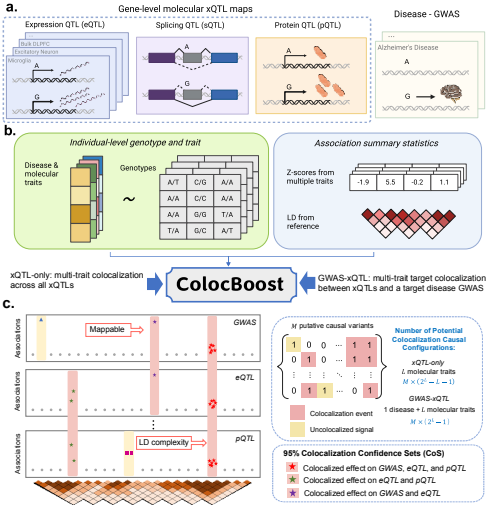
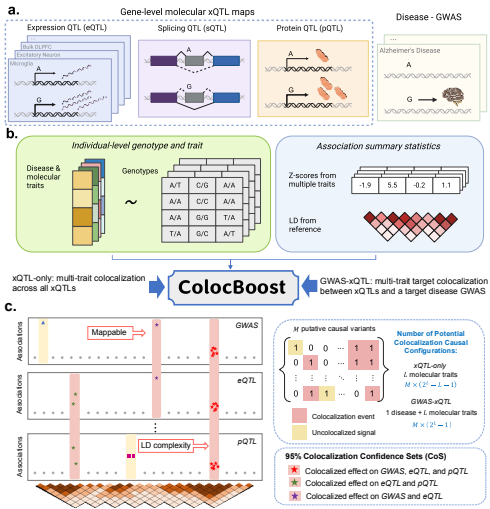



**Figure:** HyPrColoc, Foley *et al.* (2021) Nat. Comm.

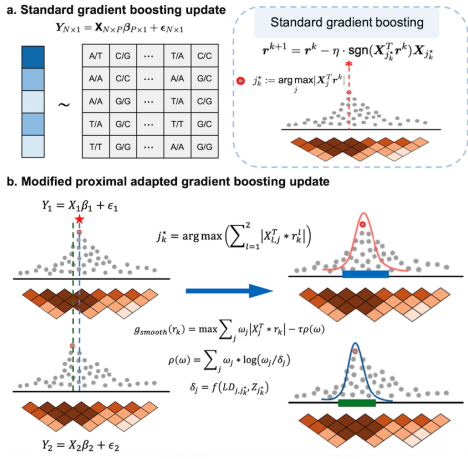
# *ColocBoost*, a new multi-trait colocalization method



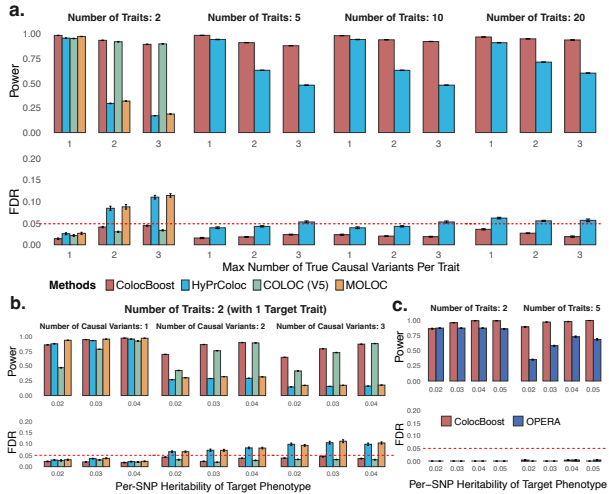
# *ColocBoost*, a new multi-trait colocalization method




## *ColocBoost* statistical model



# *ColocBoost* outperforms existing methods



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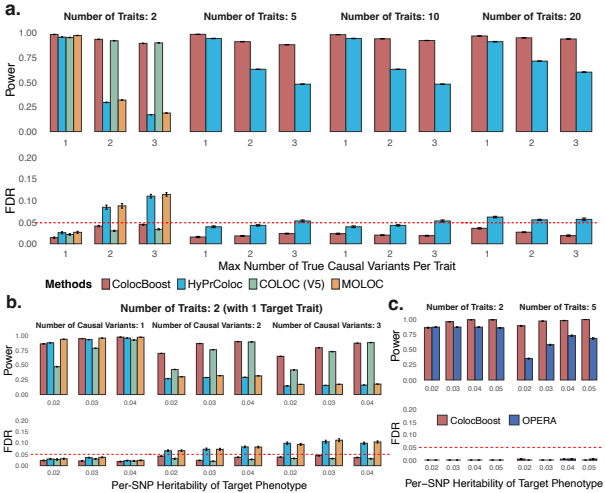
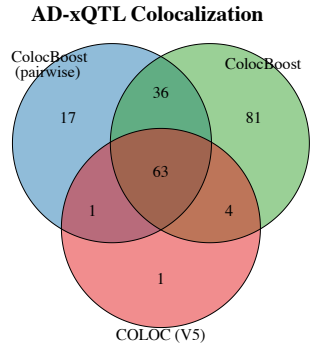
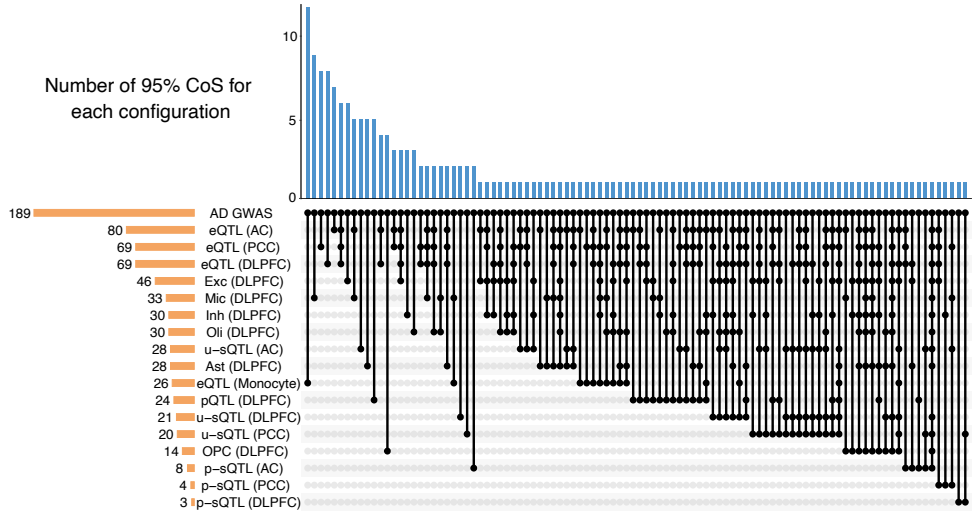


# Brain xQTL and Alzheimer's disease GWAS colocalization



# QTL-GWAS genes: transcriptome-wide association studies

# Motivation: eQTLs are enriched in GWAS signals

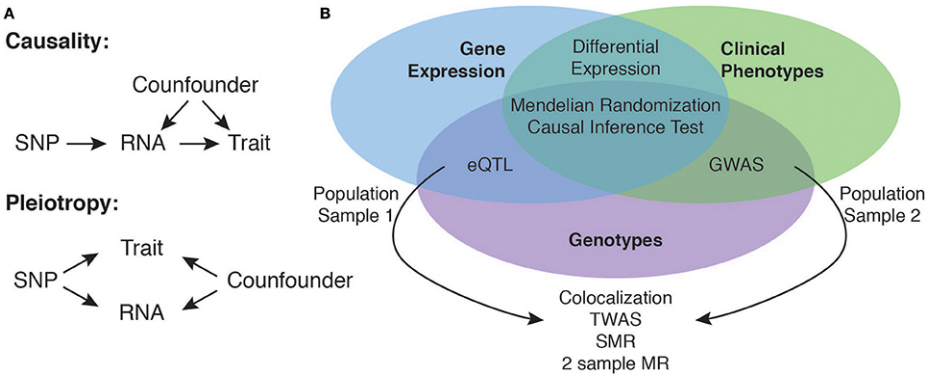



**Figure:** Heinig (2018) Front. Cardiovasc. Med.

# Transcriptome-wide association study (TWAS)

Contributions of multiple genetic variants to complex traits through their impact on molecular phenotypes

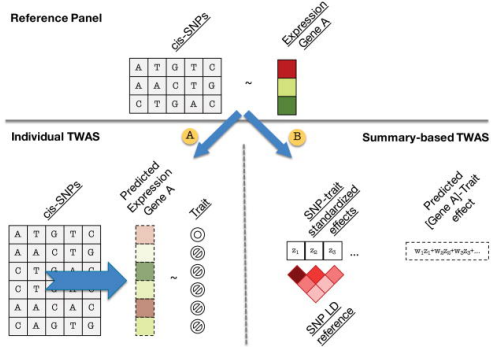
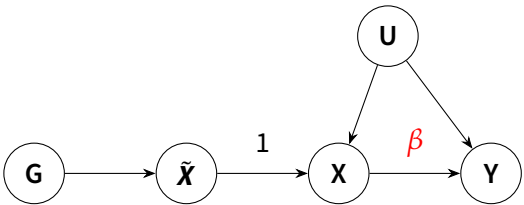



**Figure:** Gusev *et al*. (2016) Nat. Genet.

# The underlying model of TWAS




- **G**: Genetic variants.
- **X**: Gene expression.
- **X̃**: Genetic component of expression (**X** predicted from **G**).
- **Y**: Phenotype.
- **U**: Unobserved confounder.

$Y \sim \tilde{X}$ regression estimates effect $\beta$ of expression on phenotype.

# TWAS: a two-stage procedure (individual level data)

Stage 1: Build a prediction model for gene expression ($\boldsymbol{X}$) from genotypes ($\boldsymbol{G}$)

- **Model** for one gene: $X = G\boldsymbol{w} + \varepsilon_X$
- Predicted expression: $\widetilde{X} = G\widehat{\boldsymbol{w}}$

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- Predicted expression: $\widetilde{X} = G\widehat{\boldsymbol{w}}$

Stage 2: Test for association of predicted expression ($\widetilde{\boldsymbol{X}}$) with phenotype ($\boldsymbol{Y}$)

- **Model** for one gene: $Y = \widetilde{X}\beta + \varepsilon_Y$
- Test the null hypothesis: $\beta = 0$

# TWAS challenge: association vs causality

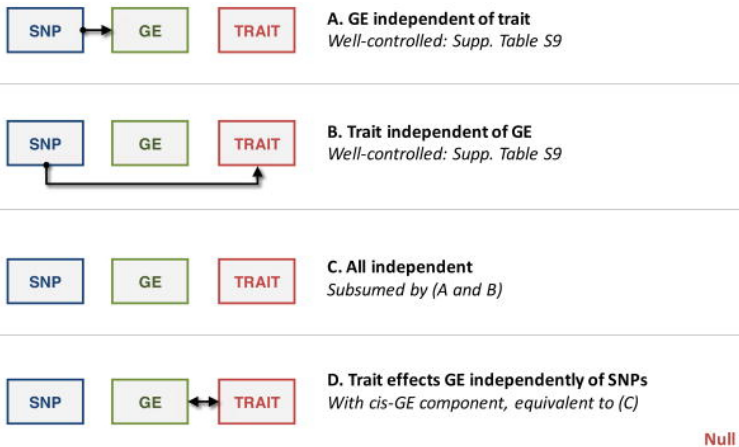



**Figure:** Gusev *et al*. (2016) Nat. Genet.

# TWAS challenge: association vs causality

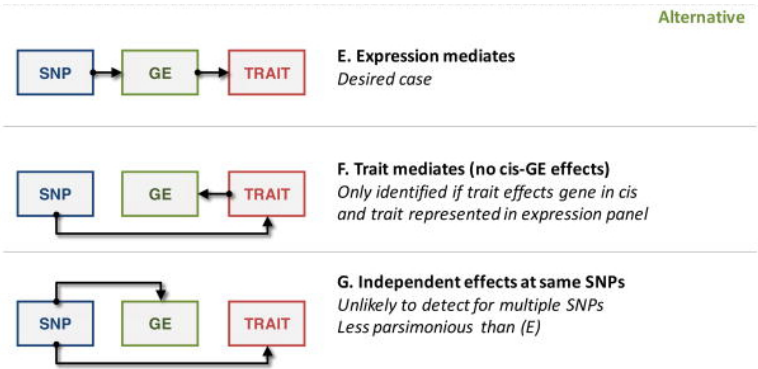



**Figure:** Gusev *et al*. (2016) Nat. Genet.

# TWAS challenge: technical considerations

Ideal TWAS setup

- Homogenous population
- Tissue and cell-type specific
- Training data-set is large and complete ($N > 200$)

But in reality

- Cross population TWAS aplications
- Multiple tissue and cell-types
- Availability of individual level data vs summary statistics

# TWAS methods review

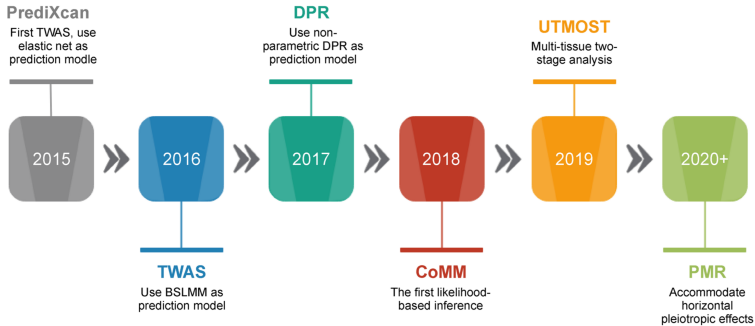
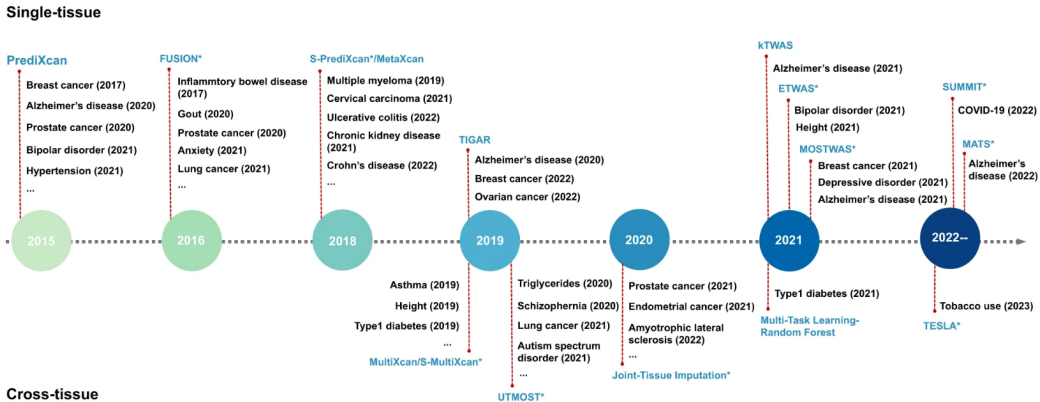



**Figure:** Zhu and Zhou *et al*. (2021) Quantitative Biology

A list of TWAS methods timeline created by Rui Dong (Leal lab, Columbia): `https://github.com/cumc/handson-tutorials/tree/main/contents/twas/20241231_TWAS_literature_review.ipynb`.

# TWAS applications review




Methods with '*' take GWAS summary statistics data as input. Exemplified applications are listed below each method.

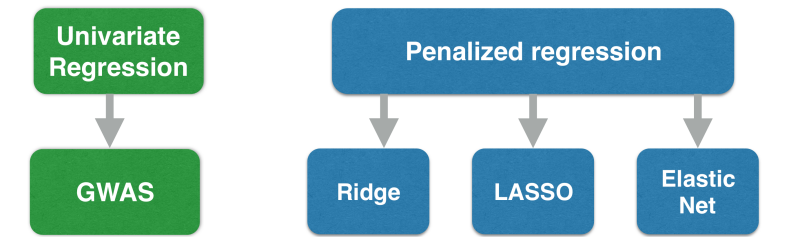
**Figure:** Mai *et al*. (2023) Communications Biology

# TWAS regression methods

Credit: slide contents on univariate TWAS methods from Dr. Haky Im
2018 HGEN 471, The University of Chicago

# Univariate TWAS methods overview

$$Y = \sum_{k=1}^{M} \beta_k X_k + \epsilon$$



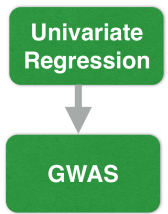
# Simple regression method

LETTERS

**Common polygenic variation contributes to risk of schizophrenia and bipolar disorder**

The International Schizophrenia Consortium*

$$Y = \sum_{k=1}^{M} \hat{\beta}_k^{\mathrm{GWAS}} X_k$$



# Ridge regression / BLUP

**REPORT**

GCTA: A Tool for Genome-wide Complex Trait Analysis

Jian Yang,[1,*] S. Hong Lee,[1] Michael E. Goddard,[2,3] and Peter M. Visscher[1]

AJHG 2011

$$Y = \sum_{k=1}^{M} \hat{\beta}_k^{\text{Ridge}} X_k$$

Penalized regression

Ridge

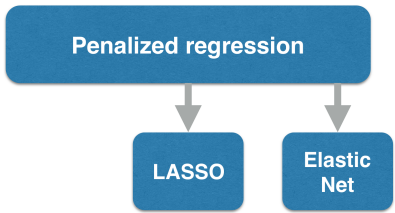
$$\|Y - \sum_k X_k \beta_k\|_2 + \lambda_2 \|\beta_2\|_2$$

# Other penalized regression

*J. R. Statist. Soc.* B (2005)
**67**, *Part* 2, *pp.* 301–320

**Regularization and variable selection via the elastic net**

Hui Zou and Trevor Hastie
*Stanford University, USA*




$$Y = \sum_{k=1}^{M} \hat{\beta}_k^{\text{E-N}} X_k$$

$$\|Y - \sum_k X_k \beta_k\|_2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta_2\|_2$$

# Bayesian variable selection regression

OPEN ACCESS Freely available online

PLOS | GENETICS

## Polygenic Modeling with Bayesian Sparse Linear Mixed Models

Xiang Zhou[1]*, Peter Carbonetto[1], Matthew Stephens[1,2]*

$$Y = \sum_{k=1}^{M} \beta_k^L X_k + \sum_{k=1}^{M} \beta_k^S X_k + \epsilon$$

$$\beta_k^L \sim N(0, \sigma_L^2)$$

$$\beta_k^S \sim N(0, \sigma_S^2)$$

## MultiBLUP: improved SNP-based prediction for complex traits

Doug Speed and David J Balding

*Genome Res.* published online June 24, 2014
Access the most recent version at doi:10.1101/gr.169375.113

# Choice of methods: cross validation

## TWAS / FUSION

### Functional Summary-based Imputation

New! RWAS (Grishin et al.) models for TCGA ATAC-seq

New! CONTENT (Thompson et al.) context-specific models for single-cell and bulk expression

New! GTEx v8 models

FUSION is a suite of tools for performing transcriptome-wide and regulome-wide association studies (TWAS and RWAS). FUSION builds predictive models of the genetic component of a functional/molecular phenotype and predicts and tests that component for association with disease using GWAS summary statistics. **The goal is to identify associations between a GWAS phenotype and a functional phenotype that was only measured in reference data**. We provide precomputed predictive models from multiple studies to facilitate this analysis.

Please cite the following manuscript for TWAS methods:

Gusev et al. "Integrative approaches for large-scale transcriptome-wide association studies" 2016 *Nature Genetics*

# Likelihood based approach

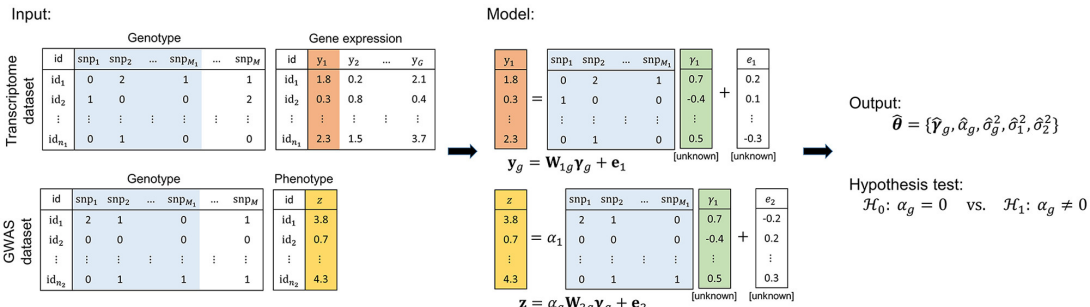
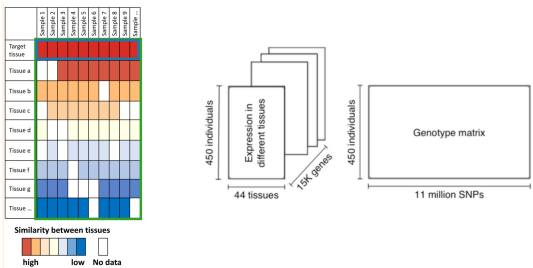



**Figure:** CoMM, Yeung *et al*. (2019)

Also see Yuan *et al*. (2022) likelihood based Mendelian Randomization

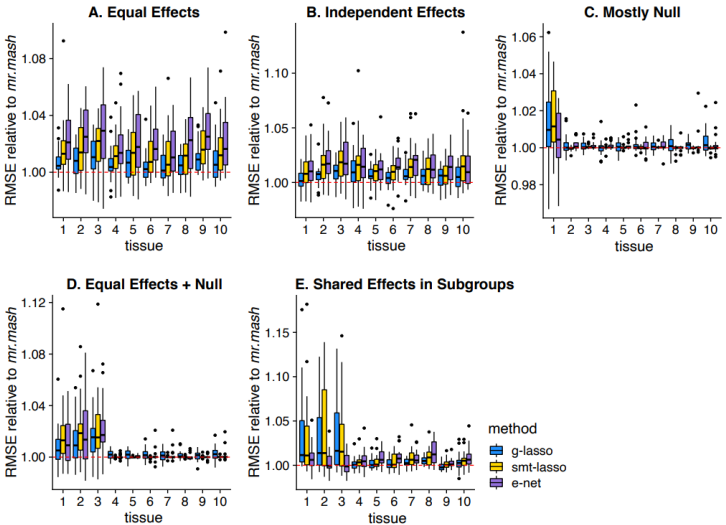
# Multivariate TWAS methods

Leverage similarity between molecular phenotypes




- UTMOST, Yu *et al*. (2019) Nature Genetics
- MR-JTI, Zhou *et al*. (2020) Nature Genetics
- mr.mash, Morgante *et al*. (2024) PLoS Genetic

# Multivariate TWAS method: mr.mash



# Prioritize genes near GWAS hits

**Step 1:** Map SNPs to genes

- Define gene boundaries: often use TSS $\pm$ 100kb window
- Consider additional regulatory regions when available

**Step 2:** Aggregate evidence

- Common test statistic: $T = \sum_{j=1}^{m} Z_j^2$ for independent $m$ SNPs
- Implementation: VEGAS (PLINK), fastBAT (GCTA), MAGMA

Note: Methods differ in how they calculate p-values for the aggregated statistics

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- Common test statistic: $T = \sum_{j=1}^{m} Z_j^2$ for independent $m$ SNPs
- Implementation: VEGAS (PLINK), fastBAT (GCTA), MAGMA

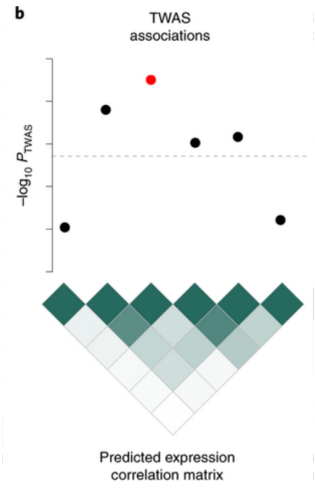
**Connection to TWAS:**

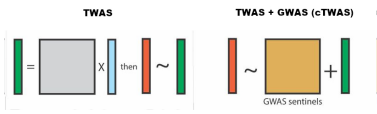
- Summary statistics based TWAS test: $Z_{\text{TWAS}} = \frac{\sum_{j=1}^{m} w_j Z_j}{\sqrt{\sum_{i=1}^{m}\sum_{j=1}^{m} w_i w_j R_{ij}}}$
  - $w_j$: **weights precomputed from molecular QTL** (*e.g.* eQTL)
  - $R_{ij}$: accounts for LD between variants $i$ and $j$
- Basic approach: Assumes $w_j = 1$ (unweighted), $R_{ij} = I$ (independent SNPs)
- Then $(Z_{\text{TWAS}})^2 = \frac{(\sum_{j=1}^{m} Z_j)^2}{\sum_{j=1}^{m} 1} \to \sum_{j=1}^{m} Z_j^2$

# Exercise: multivariate fine-mapping and TWAS

# Connections: fine-mapping, colocalization, TWAS and MR

# TWAS and fine-mapping: variable selection




Article | Published: 29 March 2019

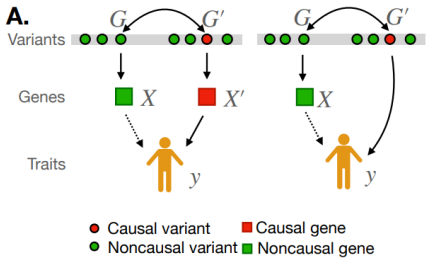
## Probabilistic fine-mapping of transcriptome-wide association studies

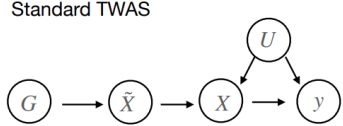
Nicholas Mancuso, Malika K. Freund, Ruth Johnson, Huwenbo Shi, Gleb Kichaev, Alexander Gusev & Bogdan Pasaniuc

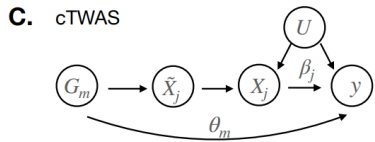
*Nature Genetics* **51**, 675–682 (2019) | Cite this article

**10k** Accesses | **115** Citations | **89** Altmetric | Metrics

# TWAS and fine-mapping: variable selection




**Model**

$$y = \sum_j \beta_j \tilde{X}_j + \sum_m \theta_m G_m + \epsilon$$

Gene effects — Variant effects

**Sparse priors**

$$\beta_j \sim \pi_G \cdot N(0, \sigma_G^2) + (1 - \pi_G) \cdot \delta_0$$

$$\theta_m \sim \pi_V \cdot N(0, \sigma_V^2) + (1 - \pi_V) \cdot \delta_0$$

**Figure:** Zhao *et al*. (2024) Nat. Genet.

# TWAS and colocalization: pleiotropy

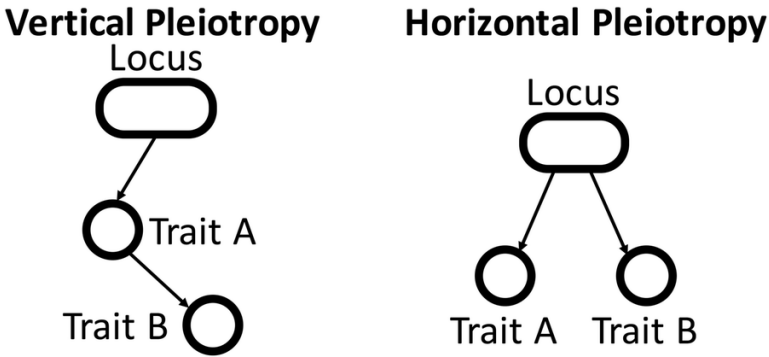
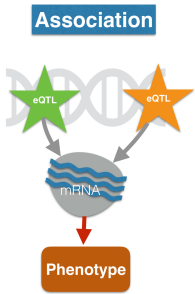
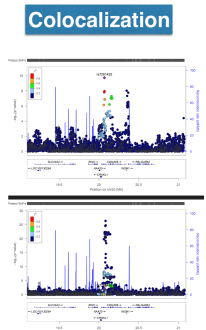



**Figure:** Jordan *et al*. (2019) Genome Biology

# TWAS + colocalization: pleiotropy




PrediXcan, SMR, FUSION

Coloc, Enloc, eCAVIAR, Sherlock

- Image credit: Haky Im
- “Locus level” colocalization: Hukku *et al*. (2022) AJHG; Okamoto *et al*. (2023) AJHG.

# TWAS and colocalization: statistical framework

$$\boldsymbol{M} = \mu_M \mathbf{1} + \boldsymbol{G}\boldsymbol{\beta}_E + \mathrm{e}_M, \mathrm{e}_M \sim \mathrm{N}\left(\boldsymbol{0}, \sigma_M^2 \boldsymbol{I}\right)$$

$$\boldsymbol{Y} = \mu_Y \mathbf{1} + \gamma \boldsymbol{M} + \boldsymbol{G}\boldsymbol{\beta}_Y + \mathrm{e}_Y, \mathrm{e}_Y \sim \mathrm{N}\left(\boldsymbol{0}, \sigma_Y^2 \boldsymbol{I}\right)$$

- "locus level", $Pr(\gamma \neq 0 | \text{Data}) \propto Pr(\gamma \neq 0) Pr(\text{Data})$
- $\Pr(\gamma \neq 0) = Pr(coloc) \times Pr(twas)$
- Data: z-score from TWAS.
- Key idea: Test $\gamma = 0$, not to estimate $\gamma$ which is Mendelian Randomization.

# TWAS and Mendelian randomization

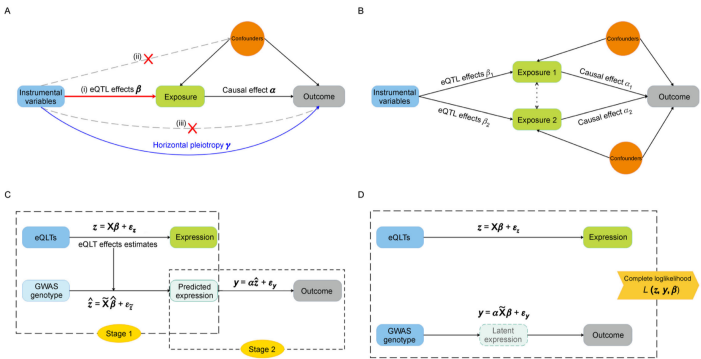



**Figure:** Zhu and Zhou (2021) Quantitative Biology

TWAS can be viewed as two-sample MR — using various IV selection methods.

# Colocalization and Mendelian randomization

- **Colocalization** answers:
  "Are gene and trait expressions caused by the same variants?"

- **Mendelian Randomization (MR)** answers:
  "Does gene expression cause traits?"

**Suggested reading:**

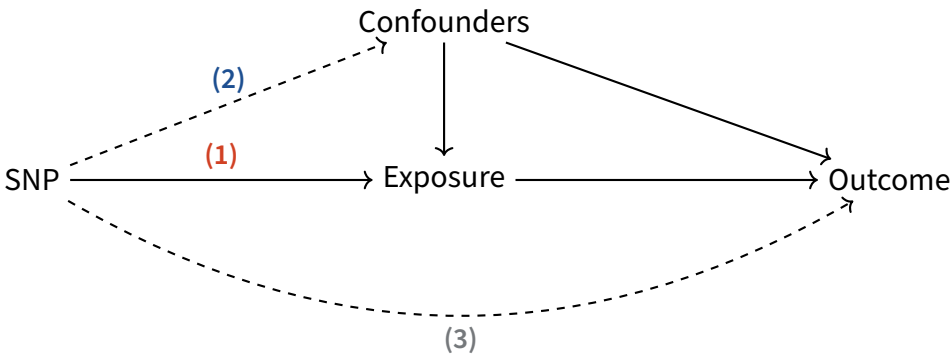
Zuber V, Grinberg NF, Gill D, Manipur I, Slob EAW, Patel A, Wallace C, Burgess S (2022). Combining evidence from Mendelian randomization and colocalization: Review and comparison of approaches. *American Journal of Human Genetics*, 109(5):767-782. doi: 10.1016/j.ajhg.2022.04.001

# Appendix: Mendelian Randomization (MR)

Model and methods outlined for reference only

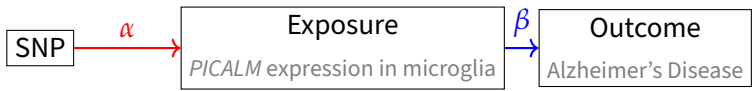
MR will be lectured in detail on Thursday

# Mendelian Randomization: 3 Core Assumptions




1. **SNP is associated with the exposure.**
2. **SNP is NOT associated with confounding variables.**
3. **SNP ONLY associated with outcome through the exposure.**

# Mendelian randomization with molecular traits as exposure




$$\text{Ratio Estimate} = \frac{\text{G-Y association}}{\text{G-X association}} = \hat{\beta}$$

# Timeline of MR methods

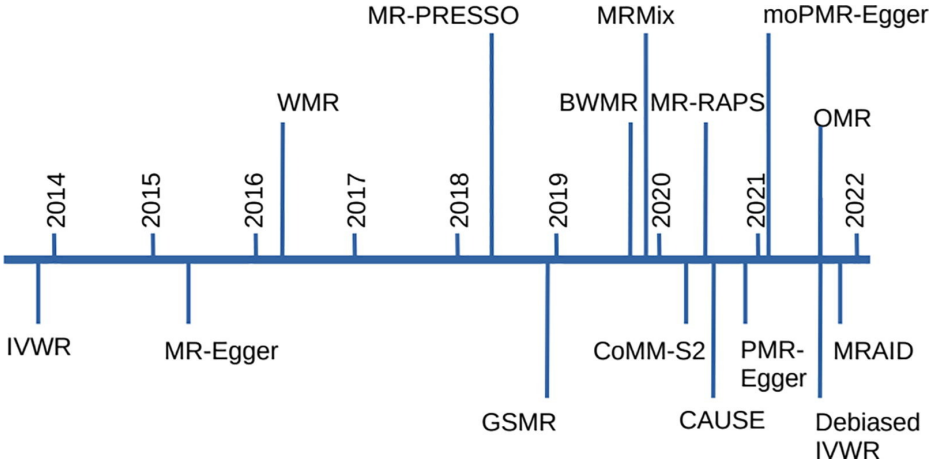



**Figure:** Boehm and Zhou (2022) Comp. Struct. Biotechnol. J.

# MRAID — MR with automated instrument determination

MRAID for individual level data

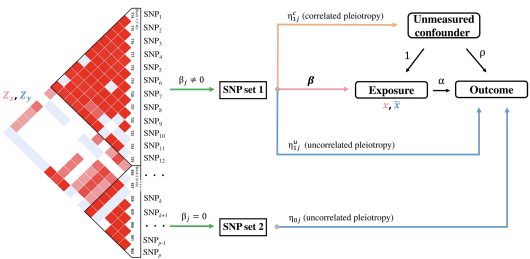



**Figure:** Yuan *et al.* (2022) Sci. Adv.

Exposure GWAS

$$\mathbf{x} = \mathbf{Z_x}\boldsymbol{\beta} + \boldsymbol{\epsilon_x}$$

Outcome GWAS

$$\tilde{\mathbf{x}} = \mathbf{Z_y}\boldsymbol{\beta} + \tilde{\boldsymbol{\epsilon}}_{\mathbf{x}}$$

$$\mathbf{y} = \underbrace{\tilde{\mathbf{x}}\boldsymbol{\alpha}}_{\substack{\text{SNP set 1}\\ \text{through}\\ \text{exposure}}} + \underbrace{\mathbf{Z_y}\boldsymbol{\eta}_0}_{\text{SNP set 2}} + \underbrace{\mathbf{Z_y}\boldsymbol{\eta}_1}_{\substack{\text{SNP set 1}\\ \text{but not}\\ \text{through}\\ \text{exposure}}} + \boldsymbol{\epsilon_y}$$

# MRAID — MR with automated instrument determination

MRAID for summary statistics

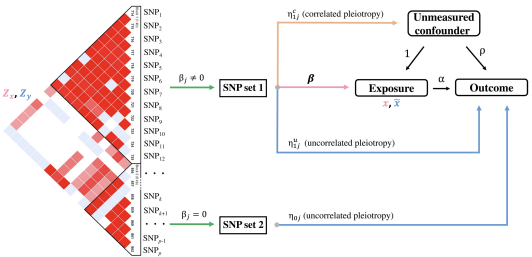



**Figure:** Yuan *et al.* (2022) Sci. Adv.

Exposure GWAS

$$\hat{\boldsymbol{\beta}}_{\mathbf{x}} = \boldsymbol{\Sigma}_1\boldsymbol{\beta} + \mathbf{e}_{\mathbf{x}}$$

Outcome GWAS

$$\hat{\boldsymbol{\beta}}_{\mathbf{y}} = \underbrace{\alpha\boldsymbol{\Sigma}_2\boldsymbol{\beta}}_{\text{SNP set 1 through exposure}} + \underbrace{\rho\boldsymbol{\Sigma}_2(\boldsymbol{\beta}\circ\mathbf{v})}_{\text{SNP set 1 through confounder Correlated plieotropy}} + \underbrace{\boldsymbol{\Sigma}_2\boldsymbol{\eta}_{\mathbf{u}}}_{\text{SNP set 1 and 2 but through neither exposure nor confounder Uncorrelated plieotropy}} + \mathbf{e}_y$$